FONA XDG

Service and Installation Manual

English

CE 0051

FONA XDG Service and Installation Manual
English Edition

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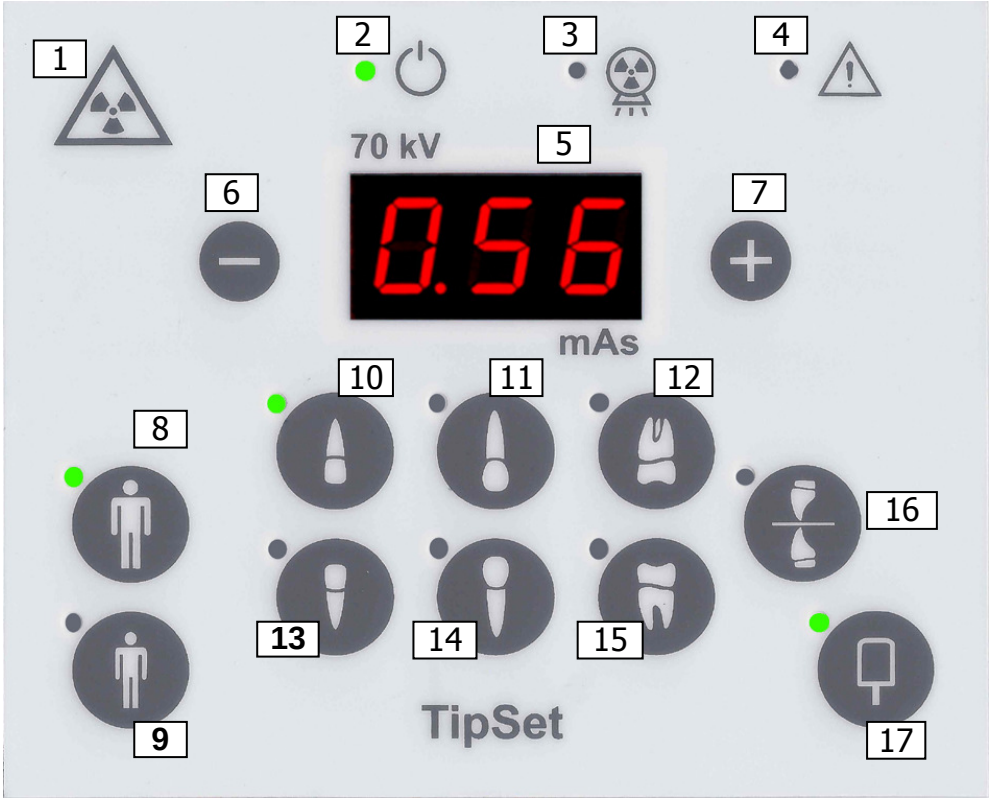
The Control Panel



1	Device for emission of ionizing radiation on request		10	Maxillary incisor
2	Indication of system turned on and ready		11	Maxillary canine or premolar
3	Irradiation		12	Maxillary molar
4	Alarm		13	Mandibular incisor
5	mAs display, the controlled technique factor		14	Mandibular canine or premolar
6	Manual decrease of controlled technique factor		15	Mandibular molar
7	Manual increase of controlled technique factor		16	Bite-wing premolar
8	Patient size adult/large		17	Digital detector in use
9	Patient size child/small		18	Radiation exposure pushbutton

EXPOSURE INDEX											
0.32	0.4	0.5	0.64	0.8	1.0	1.25	1.6	2.0	2.5	3.2	4.0
	CDR SENSOR	XIOS SENSOR		F FILM	E FILM			D FILM			
CDR Sensor by Shick Technologies Inc., XIOS Sensor by Sirona Dental Systems GmbH											

1. INTRODUCTION

1.1 Purpose

The FONA XDG X-ray Equipment is designed to fulfil the needs for high resolution intra-oral radiography in the general dental practice. The systems can be configured for wall, mobile solutions.

The Operator's Manual and the Service and Installation Manuals supplied with the system are integral part of the product. The original language of the Operator's Manual is English.

1.2 Equipment Classification

- IEC: FONA XDG is a Class I, type B equipment
- FDA: FONA XDG is a Class II medical device equipment (21 CFR 872-1800).

1.3 Applicable Standards

The FONA XDG system configurations comply with the following standards.

IEC 601-1	General requirements for safety
IEC 601-1-2	Electromagnetic compatibility
IEC 601-1-3	General requirements for radiation protection in diagnostic X-ray equipment
IEC 601-2-7	Particular requirements for the safety of high voltage generators of diagnostic X-ray generators
IEC 601-2-28	Particular requirements for the safety of X-ray source assemblies and X-ray tube assemblies for medical diagnosis
21 CFR 1020.30	Diagnostic x-ray systems and their major components
21 CFR 1020.31	Radiographic equipment
93/42/EEC	European Directive concerning medical devices (1993)

1.4 Environmental Data

Applicable ranges of temperature, humidity, and atmospheric pressure are reported here below both for operation and transport conditions.

	Ambient	Transport & Storage
Temperature	from 10 to 40 °C	from –20 to +50 °C
Relative Humidity	from 30 to 75%	10 to 90%
Pressure	from 700 to 1060 hPa	from 500 to 1060 hPa

1.5 Obligations of the Installer

Obligations of the Installer are:

- To make sure that the line voltage specified by the Manufacturer of the equipment is available and within the specified range.
- For safety reasons verify that a proper switch is available to disconnect from line voltage supply when needed during installation.
- Install and test the equipment with due diligence according to the installation instructions from the Manufacturer.
- To provide the Operating Instructions to the User.

1.6 Warning

Use the system only after proper assembly and installation as per manufacturer's instructions.

X-ray equipment produce ionizing radiation that may be harmful if not properly controlled. It is therefore recommended that the equipment be operated by trained personnel only in accordance with existing law.

Even if compliant to specifications of electromagnetic compatibility, it is recommended not to use the equipment in presence of external electromagnetic fields, such as those generated by cellular phones, which might interfere with the electronic circuits of the system.

1.7 Demonstration

In order to use of the system for demonstration purposes radiation emission has to be inhibited by disconnecting the supply cables to the X-ray head into the wall adaptor or into the timer.

Cables to be disconnected are those leaving the connection block towards the X-ray head ("out").

Make sure that the disconnected cables are properly insulated to prevent undesired contacts with live points. This task has to be done by trained personnel only to avoid the risk of electrical shock.

2. TECHNICAL DATA

2.1 System Supply

Line Voltage	115 V (from 99 V to 132 V in sub-ranges depending on THA mounted) 230 V (from 198 V to 264 V in sub-ranges depending on THA mounted)
Line Power	400 W
Line Voltage Range	108 - 132 V for type 93 250 01300, 207 - 253 V for type 93 250 01700
Line Fuse	Slow Blow: 6.3 A at 115 V, 4 A at 230 V, second fuse for two phases or cord
Line Frequency	50/60 Hz ± 1 Hz
Line resistance	≤ 0.4 Ohm at 115 V, ≤ 0.8 Ohm at 230 V

2.2 X-ray Head Assemblies

Nominal Line Voltage	120 V for type 93 250 01300, 230 V for type 93 250 01700
Nominal Line Current	6 A at 120 V for type 93 254 01300, 4 A at 230 V for type 93 254 01700
Line Voltage Range	108 - 132 V for 93 250 01300, 207 – 253 V for 93 250 01700
Anode Voltage (peak tube potential)	70 kVp ± 8% at nominal line voltage 66 kVp ± 8% at nominal line voltage – 10% 74 kVp ± 8% at nominal line voltage+ 10%
Anode Current (tube current)	3.5 mA ± 10% at nominal line voltage 3.0 mA ± 10% at nominal line voltage – 10% 4.0 mA ± 10% at nominal line voltage + 10%
Nominal Power	0.2 kW at 70 kVp, 3.5 mA, 0.1 s
X-ray Insert	CF4G070
Anode	Tungsten, angle 16° to the tube axis
Focal Spot	0.4 IEC 336
Inherent Filtration	> 2.5 mm Al/70kVp IEC 60522/1999
Duty Cycle	1/15
Radiation Leakage	< 0.1 mGy/h a 1 m (< 11.5 mR/h a 1 m)

2.3 Beam Limiting Device

Beam Limiting Device	Focus skin distance 21cm (8.27”)
	Round radiation field 6 cm (2.35”) diameter /Rectangular 3.2x4.2 cm (1.26”x1.65”)

2.4 Timer

Supply Voltage	110-120 for type 93 350 60200, 220-240 for type 93 350 60100					
Exposure factor	Time-current in mAs: 18 steps as per R10 scale from 0.21 to 11.2 mAs					
	0.21	0.28	0.35	0.42	0.56	0.70
	0.88	1.12	1.40	1.75	2.24	2.80
	3.50	4.40	5.60	7.00	8.75	11.2
Precision	± 0.04 mAs o 10% (whichever the greater) supplied at nominal line voltage					
Exposure factors	Automatic setting through tooth type selection and patient size, for use with traditional film or digital sensor, or manual setting with plus and minus keys.					
Irradiation signal	Yellow light on hand-switch and on control panel plus acoustic buzzer					
Hand-switch	Hand-switch with 3 m coiled cord, with remote mounting optional kit					
Overall size	15 cm /6” width, 24 cm /9”½ height, 9 cm /3”½ depth					

2.5 Mechanical Suspension System

Wall Adaptor	12cm/4.7” width, 24cm/9.4” height, 9cm/3.5” depth
Arm Length	Short (S): 30 cm /11.8”, Medium (M): 60 cm /23.6”, Long (L): 80 cm /31.5”, Extra Long (XL) 100 cm /39,4”
Useful Reach	Arm S: 138 cm /54.3”, M: 168 cm /66.1”, L: 188 cm /74”, XL: 208 cm /81.9”
Mobile Stand	Size 78 x 92 cm (30” ¾ x 36” ¼), Height: 186 cm /73” ¼” with scissor arm

2.6 Weights

Timer	1.7 kg / 3.7 lb
X-ray Head with BLD	6.6 kg / 14.5 lb
Round Bld	0.1 kg / 0.22 lb
Rectangular Bld	0.2 kg / 0.44 lb
Scissor Arm	11.7 kg / 25.8 lb
Support Arm	Arm S: 2.8 kg /6.2 lb , M: 4.0 kg /8.8 lb, L: 4.8 kg /10.6 lb, XL: 5.4 kg /11,9 lb
Wall Adaptor	1.3 kg / 2.9 lb
Mobile Stand	29.4 kg / 64.8 lb

3. ASSEMBLY AND INSTALLATION

3.1 Wall Mounted Systems

3.1.1 Installation Options

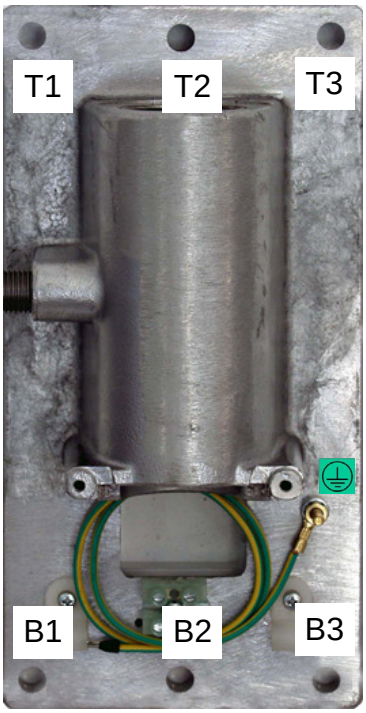
The timer box can be mounted together with the X-ray head on the right of the wall adaptor or can be placed remote either inside or outside the examination room. The release hand-switch can be directly connected to the timer box or placed remote making use of specific mounting kit including 10 m cable. A door contact and/or a second exposure button can be added via the internal socket on the timer board.

3.1.2 Unpacking

Unpack the components of the system and check the following:

A Each item is in good conditions and was not damaged during transportation.
B All the items for the desired system configuration are available.
C The line voltage on the labels of timer and X-ray head are corresponding to the local line voltage.

3.1.3 Structural Requirements



The wall adaptor has to be mounted in a convenient position on left or right side of the chair or on back wall (head of the patient).

The maximum useful reach is of 208 cm (81.9”) from the wall when using an 100 cm (39.4”) support arm.

The wall adaptor can be mounted with 2, 4, or 6 bolts, depending on wall quality.

A Two bolts only (top and bottom central holes – T2, B2) are used when there is a solid slim column (e.g. iron mounting) with weak sides (e.g. wooden wall). Considering the requested safety factor, the top bolt has to withstand a load of 6000 N, comprehensive of a safety factor, i.e. about 1350 lbs or 612 kg. Proper screw to be selected for a solid connection to the wall. Classes ISO 8.8 (M 8, M 8x1, M 8x1.25) or SAE Grade 5 (5/16” 18UNC, 5/16” 24 UNF) are recommended.

B Four bolts, two on top sides (T1, T3) and two on bottom sides (B1, B3) is the regular mounting for solid (concrete) wall, but also on large metal plate. Considering the requested safety factor, each bolt at top has to withstand a load of 3000 N, comprehensive of a safety factor, i.e. about 675 lbs or 306 kg.
Proper expansion screw to be selected for a solid connection to the concrete wall; the permissible load of each screw has to be greater than 308 lbs (about 140 kg).
- On solid concrete use heavy duty metal anchors.
- On hollow bricks use injection chemical fixing.

C Six bolts, three on top (T1, T2, T3) and three at bottom (B1, B2, B3) are required when the wall is not solid enough and the load has to be distributed on more points. Considering the requested safety factor, each bolt at top has to withstand a load of 2000 N, comprehensive of a safety factor, i.e. about 450 lbs or 204 kg.

In case the wall is not in condition to withstand the indicated load, corrective actions can be evaluated by adoption of reinforcing plates:

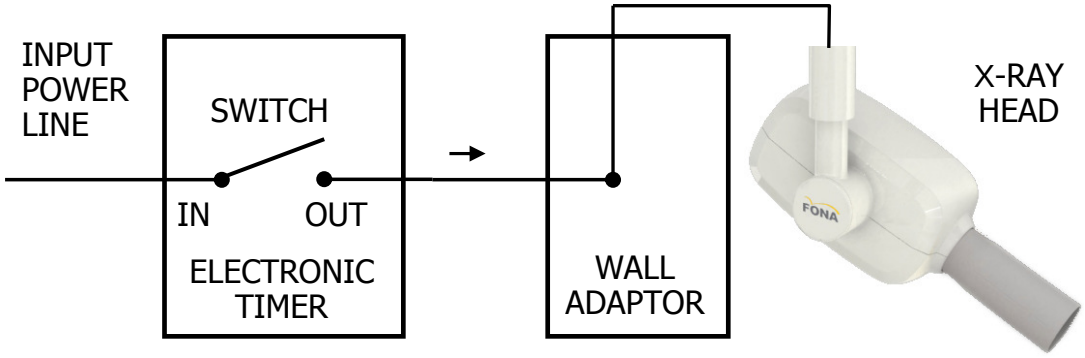
D Large plate to fit vertical supports at 16” distance, with 4 mounting holes and one cable opening for the wall mount in the middle.
E In case of a thin (wooden) wall not solid enough, the use of a steel counter plate 2 mm thick can be the solution. Make sure that the wall is solid enough to carry the load.
F The use of two reinforcing steel plates of about 4 times the surface of the wall adaptor, one by each side of the wall, can help when a single plate looks not adequate; additional bolts have to be used to hold together the two plates.
Make sure that the wall is solid enough to carry the load.

Recommended Bolts		
Diameter	Class	Core Section mm^2
M 8X1.25	ISO 8.8	36.6
M 8X1	ISO 8.8	39.2
5/16” – 18 UNC	SAE- Grade 5	33.8
5/16 –24 UNF	SAE- Grade 5	37.41

INSUFFICIENT WALL OR HARDWARE STRENGHT MAY CAUSE THE WALL ADAPTOR TO PULL OUT FROM THE WALL AND THE FULL SYSTEM TO FALL ON TO THE PATIENT OR THE OPERATOR CAUSING INJURIES.

3.1.4 Electrical Requirements

The power line cable must be connected to the input terminals (IN) of the timer to supply the timer itself and to make available power for the X-ray head at the output terminals (OUT), upon request by the operator (via the hand switch). Power to the X-ray head is thus controlled by the timer (acting as a SWITCH).



The cables (2 poles plus ground) to connect the power line to the timer and the timer to the wall adaptor are not provided. Cables with large section conductors have minimum electrical resistance and voltage drop.

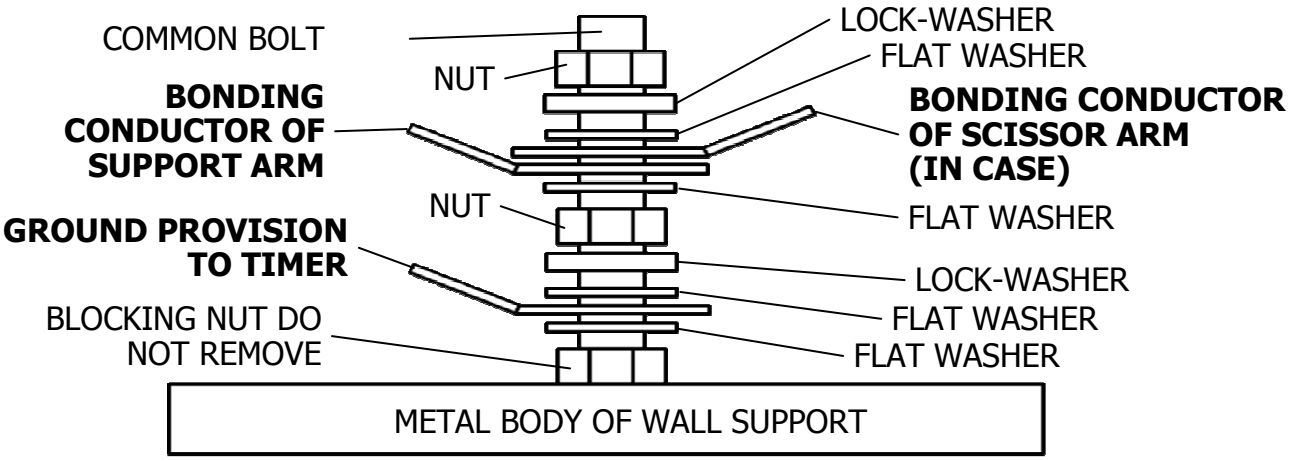
Wire Type	Cross Section Area	Resistance/m	m/0.05 Ohm
AWG 14	2.08 mm^2	0.00829 Ohm/m	6.0 m
AWG 12	3.31 mm^2	0.00521 Ohm/m	9.6 m

The resistance between the ground provision in the connecting block of the timer and the metal body of the tube housing assembly has not to exceed 0.1 Ohm. Proper section of the yellow green wire to be used in case remote mounting of the timer.

The wall adaptor is provided with a common bolt for bonding and grounding blocked in place with a nut which has not to be removed.

The grounding conductors to be connected as indicated below.

- First the conductor to ground provision in the connecting block of the timer (flat washer, ground provision, flat washer, lock-washer, nut).
- Second the bonding conductor of support arm plus, in case, the one of the cable in the scissor arm (flat washer, bonding conductors, flat washer, lock-washer, nut).



3.1.5 Mounting and Connecting Sequence

Step 1	☐	Mount the Wall Adaptor	Step 7	☐	Connect the Hand-Switch
Step 2	☐	Mount the Timer	Step 8	☐	Optional Remote Hand-Switch
Step 3	☐	Mount the Support Arm	Step 9	☐	Mount and connect X-ray head
Step 4	☐	Mount the Scissor Arm	Step 10	☐	Mount the collimator
Step 5	☐	Connect the Wall Adaptor	Step 11	☐	Final Tuning and Set-Up
Step 6	☐	Connect the Timer			

3.1.6 Mounting the Wall Adaptor

A Take away the plastic cover removing the screws under the logo label.

B Use the Wall Adaptor plate or a template to mark the holes on the wall.
Please note that the timer can be mounted close to the Wall Adaptor on the right side or in a remote position. Cabling should have been laid out accordingly.

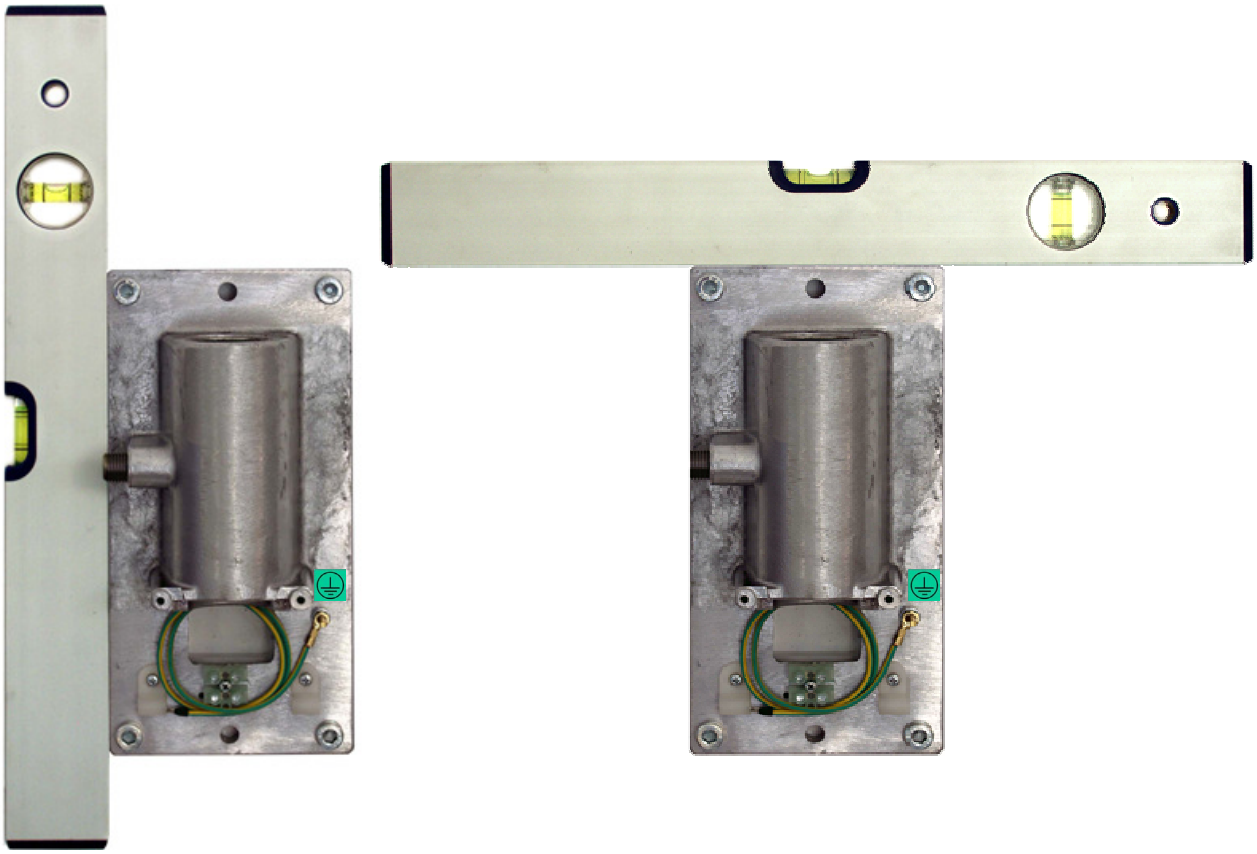
C Make the holes in the wall according to the applicable type of mounting with two, four bolts, or six bolts, with or without reinforcing plate.

D Slide the logo-strip out from the plastic cover to access the screws and remove it from the metal frame.

E Mount the metal frame on the wall using proper heavy-duty metal anchors and make sure that the power cable enters from behind.

F Secure it to the wall ensuring it is leveled. Improper leveling might cause the arm to move and swing out of position.

G Perform wire connection and final set-up complying with recommended sequence of actions reported in the following.

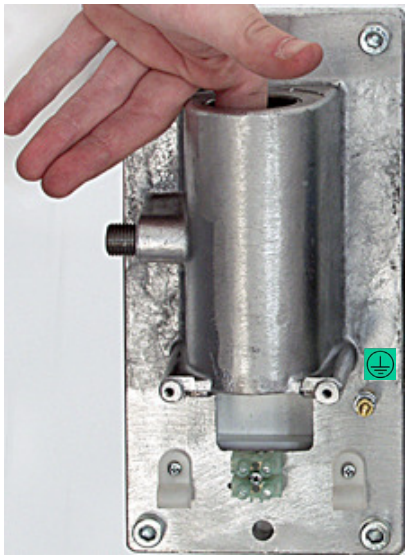


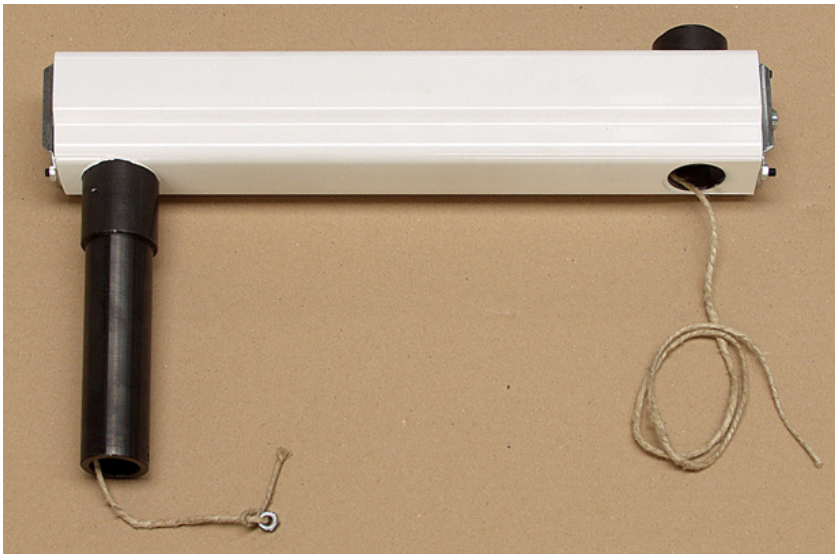
3.1.7 Mounting the Timer

A Use the mounting plate or the template to mark the holes on the wall.
B Remove the plastic cover after having taken away the blocking screws behind the logo strip. Pay attention when disconnecting the flat cable of the control panel from the control board.
C Drill the four holes in the wall and secure the timer making sure it is leveled. Cabling should have been laid out in accordance to mounting requirements.
D Perform wire connection and final set-up complying with recommended sequence of actions reported in the following.

3.1.8 Mounting the Support Arm

A Unpack the support arm and check for completeness of parts.
B Do not remove the rope to pull the cable of the Scissor Arm to the Wall Adaptor through the Support Arm.
C Push back the cylinder of the side friction not to interfere, lightly grease the shaft of the extension arm and insert it into the wall adaptor.
D Mount the rotation end-stop spacer and close the side friction without tightening.





3.1.9 Mounting the Scissor Arm

WARNING. THE SPRINGS IN THE SCISSOR ARM MAY CAUSE INJURY TO THE INSTALLER AS WELL AS DAMAGE TO THE ARM ITSELF IF NOT HANDLED PROPERLY. DO NOT REMOVE THE BINDING STRING UNTIL WHEN NECESSARY

A Light grease the shaft at the base of the arm and make sure the ring spacer is in place; push back the cylinder of the friction into the support arm not to interfere during the insertion of the shaft.

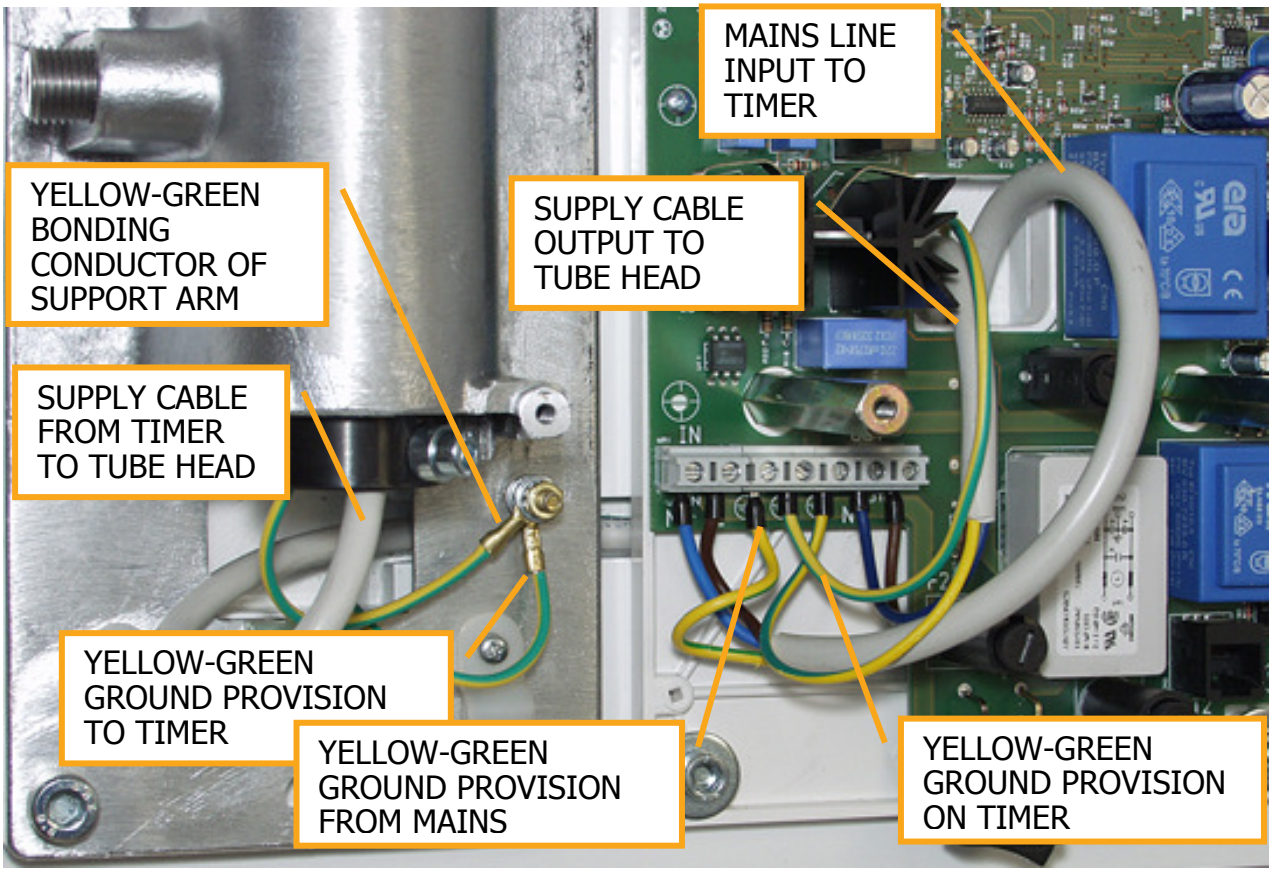
B Secure the trailing rope to the end of the electrical cable of the Scissor Arm.

C Pull the other end of the trailing rope to drive the electrical cable of the Scissor Arm through the Support Arm out into the Wall Adaptor.

D Hold the opening section of the Scissor Arm, carefully remove the safety string and allow the arm to open slowly, away from people.

E Perform final set-up complying with recommended sequence of actions reported in the following.

3.1.10 Connecting the Wall Adaptor with Timer at side



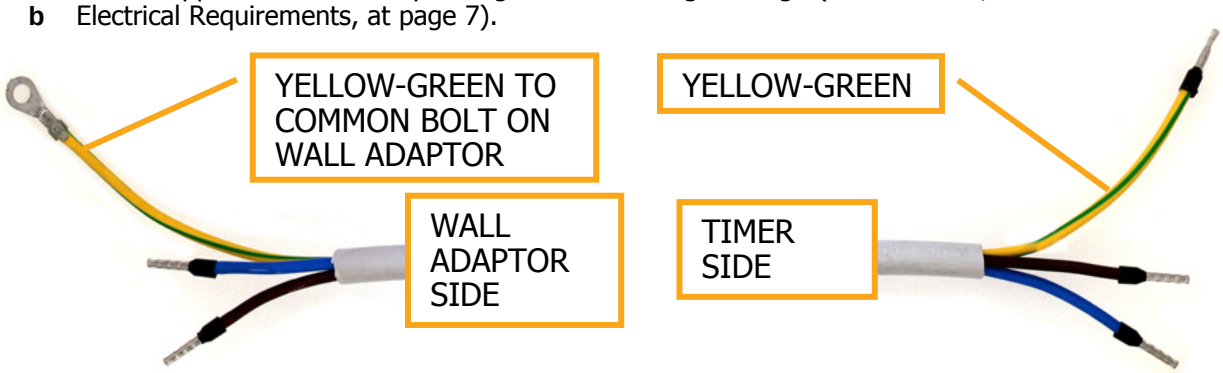
A Pass the conductor of the grounding provision (already connected to the common bolt) into the timer for connection to the central ground point in the connecting block (protective earthing).
B Connect the grounding provision of the extension arm to the common bolt on the wall plate (flat washer, ground provision, lock-washer, nut).
C Pass through the supply cable of the arm into the timer for connection to the OUT positions (supply cable output to X-ray head).

3.1.11 Connecting the Wall Adaptor with Timer remote

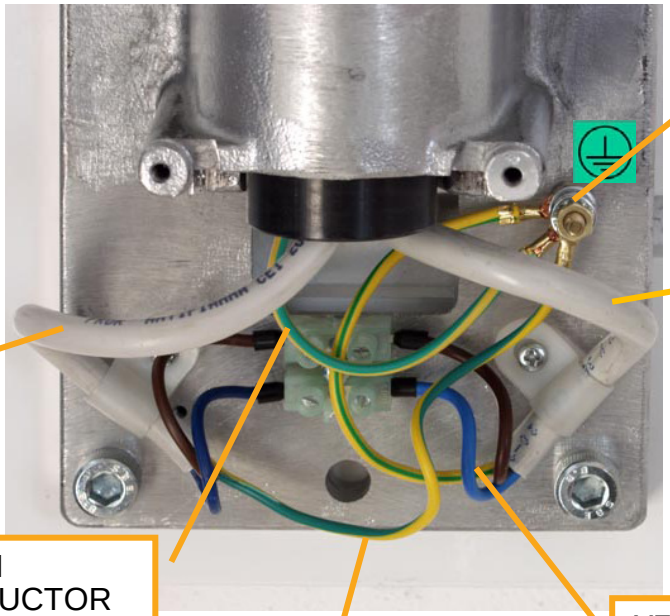
A In case the connection to the mains is done starting from the wall adaptor a 5 position connection block has to be mounted. This approach is recommended allowing for a lower ground resistance.
- **a** The ground wire of the mains has to be connected in first position on the common bolt and blocked with a nut.
- **b** The mains line goes to the remote timer and back through the connection block to the X-ray head.
- **c** The other grounding provisions for support arm and of cable returning from the timer to be connected to the common bolt and blocked with a second nut.
- **d** The ground resistance to consider is actually the one of the cable in the scissor arm.

B In case the connection to the mains is done through the timer box, the ground resistance to consider includes the wire from the timer to the wall adaptor.
- **a** Prepare the terminations of the cable for remote connection of timer with tips to fit the connecting blocks in timer and connecting blocks plus the common bolt for grounding in wall adaptor.

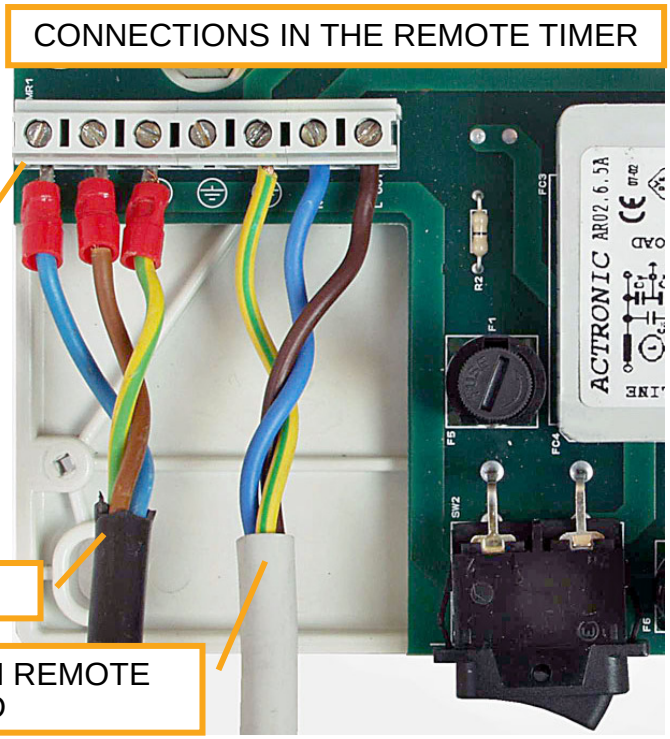
Make sure the copper section of the yellow green wire is large enough (see section 0,
- **b** Electrical Requirements, at page 7).



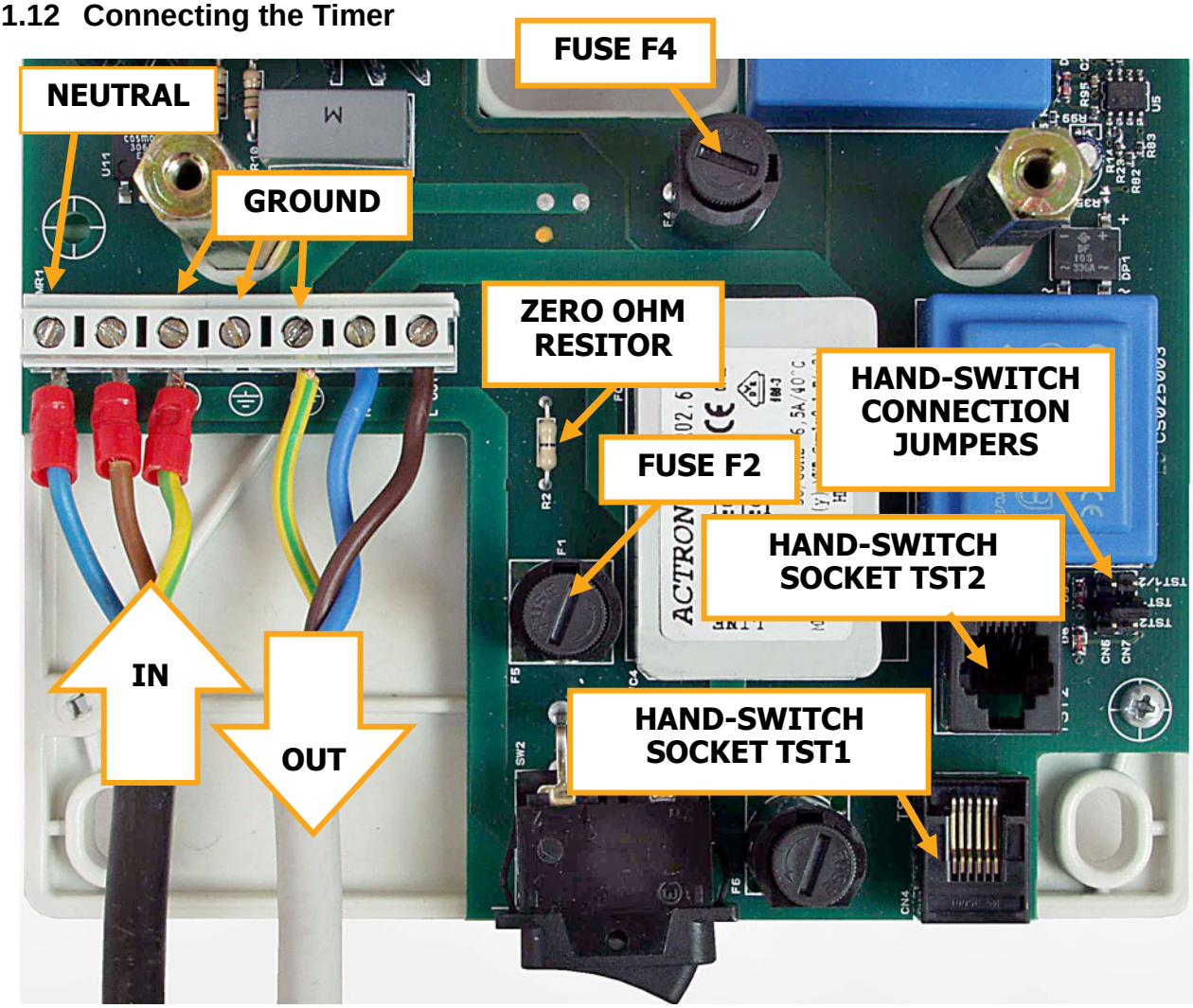
C Remove from the common bolt for bonding and grounding on the wall adaptor the existing cable for ground provision.



D Connect the cable for remote connection of the timer by placing the ground wire as first conductor on the common bolt for bonding and grounding; block with washers and nut.
E Place the phase and neutral wires to the connecting block.
F Connect to the common bolt for bonding and grounding the bonding conductor of the extension arm and the one of the cable of the arm.
G Connect the phase and neutral conductors of the supply cable in the arm to the connecting block.
See section 0,
H Electrical Requirements, at page 7 to determine cable length and maximum distance.



3.1.12 Connecting the Timer



A Turn-off the voltage supply line. Connect the three wires from the voltage supply to the terminal block (IN on the left) but do not connect any outgoing wire (OUT on the right) to Wall Adaptor.

B Make sure that the "live" conductor is the "hot" one:
- **a** Connect an AC voltmeter or a test light between the block terminal "live in" and "ground".
- **b** Turn line voltage supply "ON". If full line voltage is measured (test lamp lights) the wiring is correct. If not turn line voltage supply "OFF", reverse "live in" and "neutral in" wires and repeat the measurement; eventually the full line voltage should be read between "line" and "ground" (test lamp lights).
- **c** Make sure no voltage is read between "neutral in" and "ground"; if not check line voltage distribution.

C Test the timer for full functionality keeping the output cables disconnected not to supply the load.

D Connect the load.
- **a** Turn OFF the line voltage supply line,
- **b** Connect the three wires of additional cable outgoing to the Wall Adaptor, Arm, Tube Housing Assembly. Live and neutral wires to the arm (X-ray head) can be interchanged.

Fuses of FONA XDG Timer		
Ref.	115 V	230 V
F1	6.3 AT 5x20	4 AT 5x20
F2	6.3 AT 5x20	4 AT 5x20
F3	315 mAT 5x20	315 mAT 5x20
F4	80 mAT 5x20	50 mAT 5x20

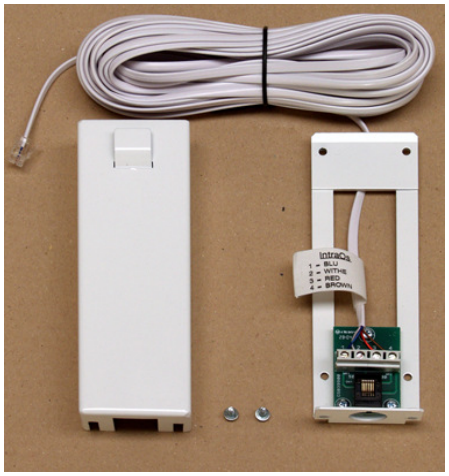
3.1.13 Connecting the Hand-Switch

A The hand-switch is provided with a 3 m coiled cord to be plugged one side in the hand-switch, the other side in the lower left corner of the timer's board, either externally (connector TST1) or internally to the timer (connector TST2). Internal connection is for remote hand-switch or for additional control safety switches.

B Jumpers are available to activate properly the hand-switch connection:
- **a** Position TST1 to enable external socketTST1,
- **b** Position TST2 to enable the internal socket TST2,
- **c** Position TST1/2 f to enable the internal and the external sockets.

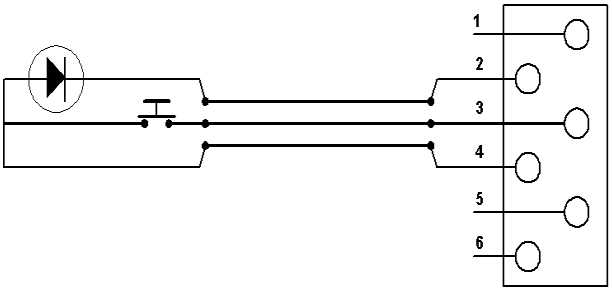
3.1.14 Optional Remote Hand-Switch

The hand-switch can be remotely mounted by making use of the optional kit made of a remote box (wall holder) and connection cable 10 m long.

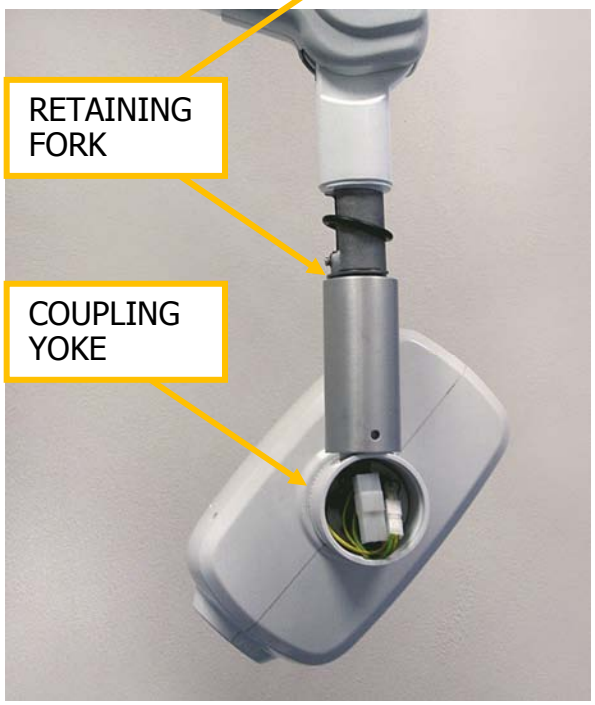


The connection of the wires is here indicated.

The wires of the switch are number 3 and 4 (those in the middle of the connector).



3.1.15 Mounting and Connecting the X-ray Head



A Remove the handgrip pipe from the arm and place it onto the tube-head shaft of the coupling yoke.

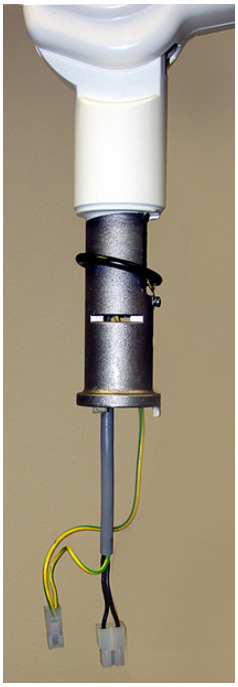
B Lubricate the shaft, insert it into the arm passing the cables into the coupling yoke, then hold it in place with the retaining fork.

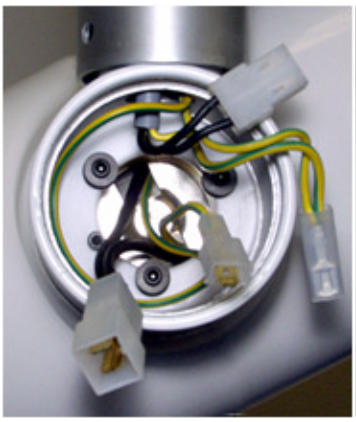
C Attach the connectors of the supply and grounding wires and push them into the side opening of the coupling yoke.

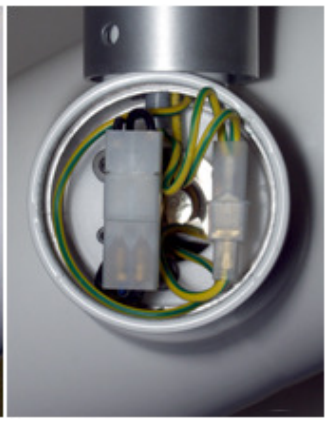
D Pull down the O-ring and push-up the handgrip pipe, then fix it in place with the screw at bottom.

E Verify smooth movement of the tube head around its vertical and horizontal axis. Tune side friction if necessary referring to section 4.4, Maintenance of X-ray Head, at page 22 for details.

F Mount the cover on the yoke opening eventually.





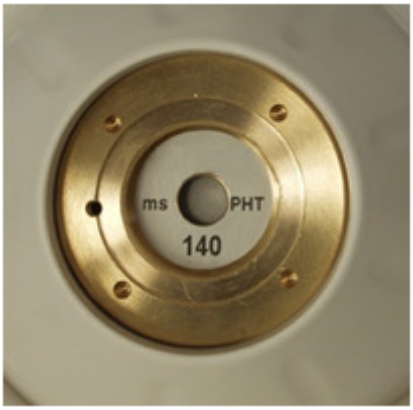


3.1.16 Beam Limiting Device

A Mount the beam limiting device on the tube-head and lock it close with a clock-wise rotation.

3.1.17 Final Tuning and Set-Up

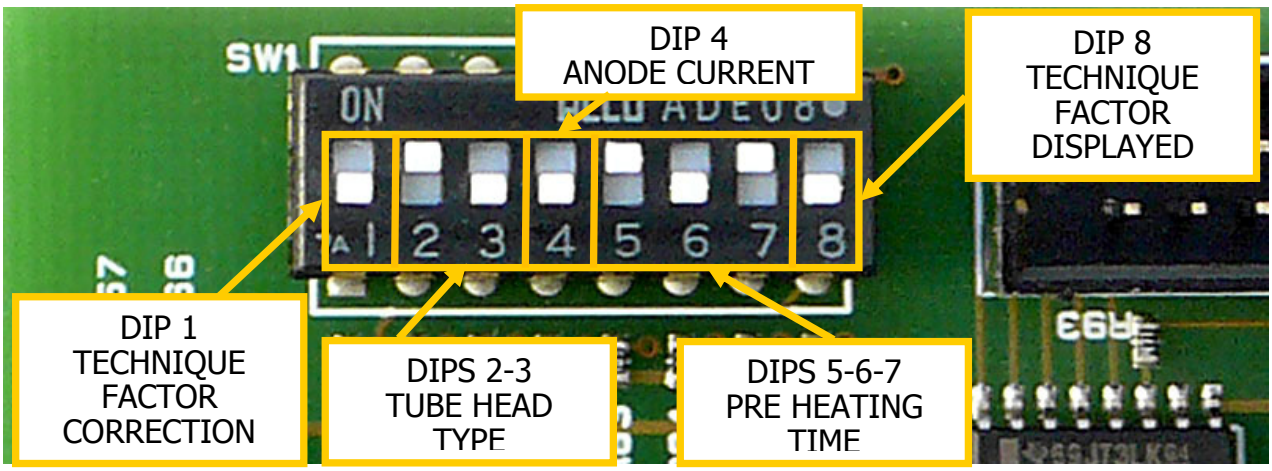
A Remove the line voltage fuse(s) from the timer (Fuse 1 and Fuse 2).
B Tune friction screw in Wall Adaptor for rotation of Support Arm (see section 4.1, Maintenance of Wall Adaptor at page 20 for details).
C Tune friction screw in Support Arm for rotation of Scissor Arm around it vertical axis (see section 4.2, Maintenance of Support Arm at page 20).
D The friction points in the scissor arm and those for the rotation of the X-ray head around its horizontal axis are tuned in factory. Should you need to repeat the procedures refer to sections 4.3, Maintenance of Scissor Arm, at page 21 and 4.4, Maintenance of X-ray Head, at page 22, respectively.
E Make sure the Timer switch is in the "OFF" position.
F Set the dip switches SW2 as indicated in the following to define:
- **a** Enable/disable function for correction of exposure in presence of line voltage fluctuations.
- **b** The nominal line voltage of the X-ray head (see label).
- **c** Set dip switch 4 to the anode current set in the X-ray head label, which is 3.5 mA for FONA XDG.
- **d** Set the pre-heating time according to type of X-ray insert used in the X-ray head assembly; the measured value is marked on the X-ray head.
- **e** Select the technique factor on display as mAs (always mAs for the timer of FONA XDG).



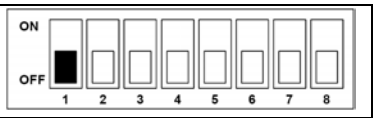
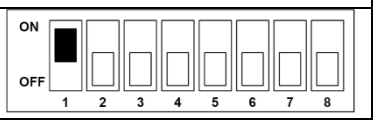
G Put back the line voltage fuse(s).
H Put back all plastic covers and logo strips.
I Connect the system to the line voltage by switching the general line voltage switch on and/or plug the power cord into the wall socket.
J Switch the timer "ON".

The system is now ready for functional checks.

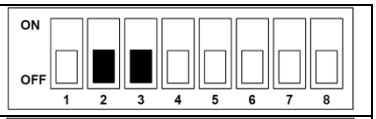
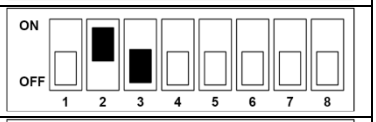
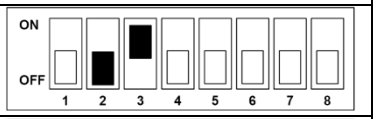
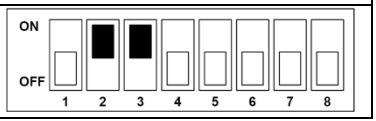
3.2 Dip Switch Setting



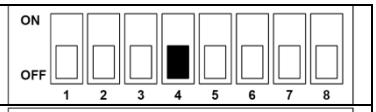
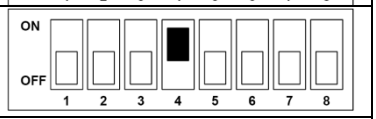
DIP 1
Technique Factor Correction

Disabled	
Enabled	

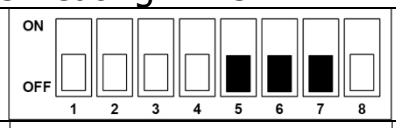
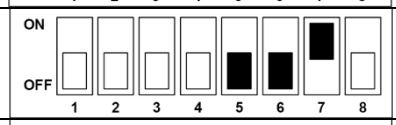
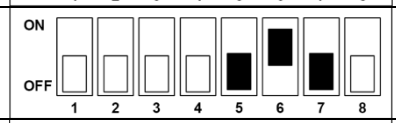
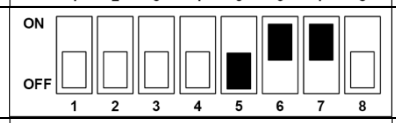
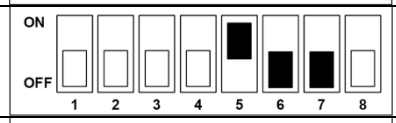
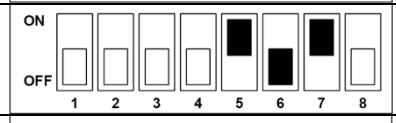
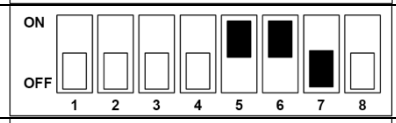
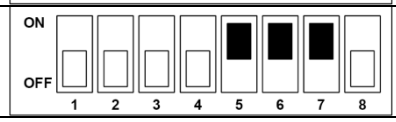
DIPS 2- 3
Tube Head Type

110 or 220 VAC	
115 or 230 VAC	
120 or 240 VAC	
127 or 250 VAC	

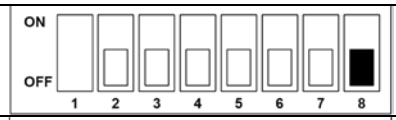
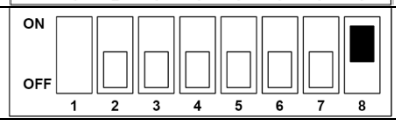
DIP 4
Tube Head Anode Current

3.5 mA	
7 mA	

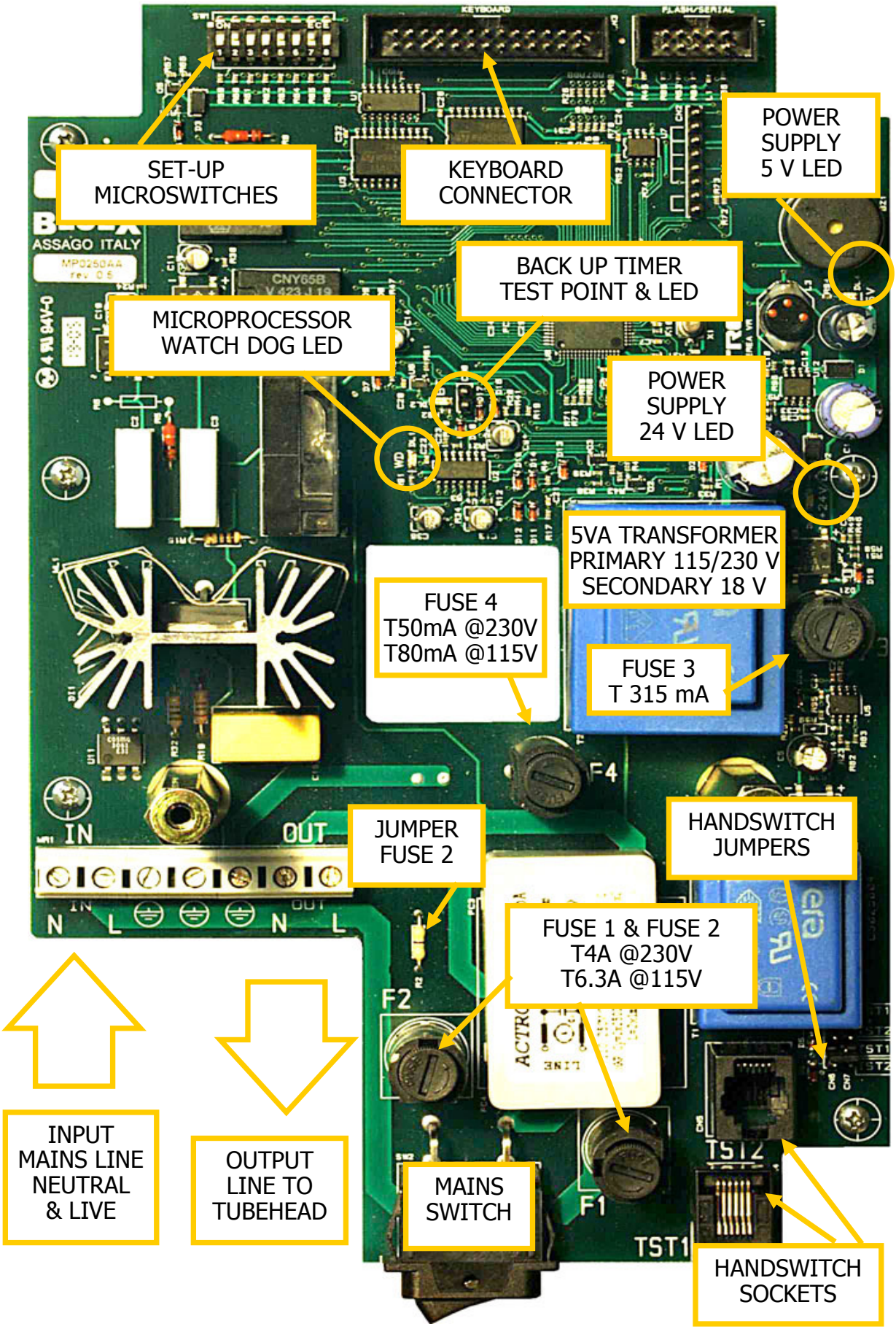
DIPS 5 – 6 – 7
Pre Heating Time

100 ms	
120 ms	
140 ms	
160 ms	
180 ms	
200 ms	
220 ms	
240 ms	

DIP 8
Technique Factor Displayed

s	
mAs	

3.3 Layout Power Board Timer



3.4 Mobile Systems

3.4.1 Unpacking

Unpack the components of the system and check the following:

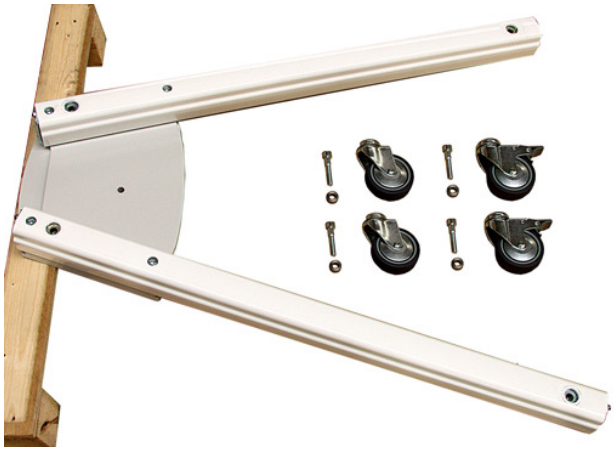
A Each item is in good conditions and was not damaged during transportation.
B All the items for the desired system configuration are available.
C The line voltage on the labels of timer and X-ray head correspond to the existing local line voltage.

3.4.2 Room Preparation

Make sure that a wall socket is available close to the dental chair in reach of the mobile X-ray equipment.
Such a socket has to be provided with connection to the protective grounding.
The line voltage cable can come without plug to be added according to the local standard.

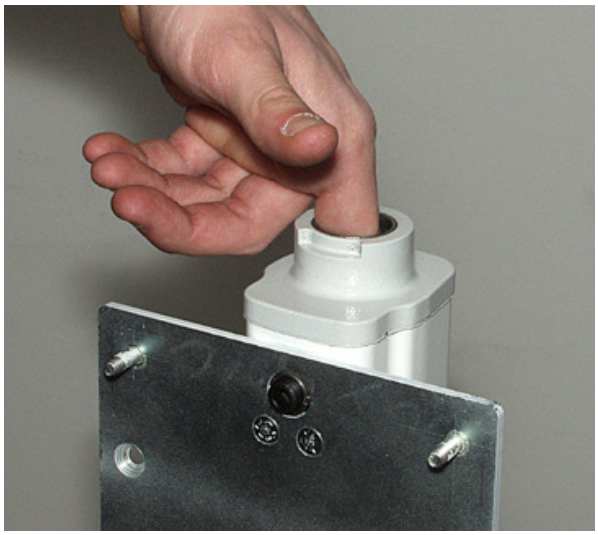
3.4.3 Mounting and Connecting Sequence

Step 1 ☐ Assemble the Mobile Stand
Step 2 ☐ Mount the Scissor Arm
Step 3 ☐ Mount the Scissor Arm
Step 4 ☐ Mount the Timer
Step 5 ☐ Connect the Line Voltage Cable
Step 6 ☐ Connect the Timer
Step 7 ☐ Activate second Fuse
Step 8 ☐ Mount and connect X-ray head
Step 9 ☐ Mount the collimator
Step 10 ☐ Final Tuning and Set-Up



3.4.4 Assembling the Mobile Stand

A Mount the legs below the central base.
B Mount rear wheels and front wheels with brakes.
C Attach the pole with timer plate.
D Complete the stand with cable holder and handles



3.4.5 Mounting the Arm

A Loosen the friction and push back the cylinder of the same not to interfere during shaft insertion.
B Insert the arm. Refer to section 3.1.9 at page 10 for details.

3.5 Mounting the Timer

The timer is blocked in place with four bolts.

3.5.1 Connecting the Line voltage cable

A The line voltage cable for the mobile has to be completed with local plug.

B The power supply cord bonding conductor has to be blocked (first) to the common bolt for bonding and grounding using lock-washer and nut.

C The grounding wire for the timer is then placed on the same common bolt and blocked (second) with lock- washer and nut.

D Connect the grounding wire for the timer to the ground provision on the terminal block of the timer and the two line wires of the power cord to the input points of the terminal block of the timer.

3.5.2 Connecting the Timer

For the connection of the output cable to the timer please refer to section 3.1.11 at page 13.

3.5.3 Activation of Fuse 2

Units provided with supply cord and plug must be equipped with fuses on both line phases. The activation on the second fuse is done by cutting the Jumper Fuse 2 (zero Ohm resistor) close to Fuse F2. See position on layout in section 0 at page 17.

3.5.4 Mounting and Connecting the X-ray Head

Refer to section 3.1.15 at page 14.

3.5.5 Mounting the Beam Limiting Device

Refer to section 3.1.16 at page 15.

3.5.6 Final Tuning and Set-Up

Refer to section 3.1.17 at page 15.

4. MAINTENANCE

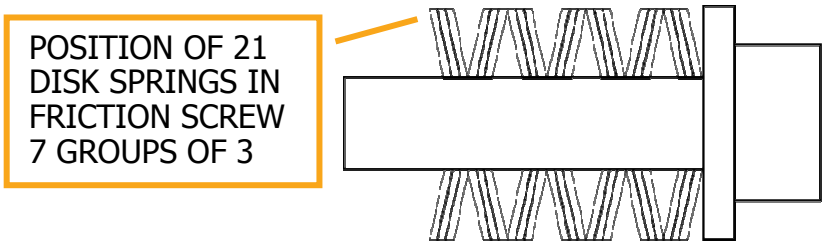
Here in the following the list of recommended maintenance actions to keep the FONA XDG system in compliance with the Performance Standard.
It is the responsibility of the User to maintain the equipment in compliance with the standard. Failure of the User to properly maintain the equipment may relieve the Manufacturer, or its Agent, from responsibility for any injury, damage or non compliance which may result.

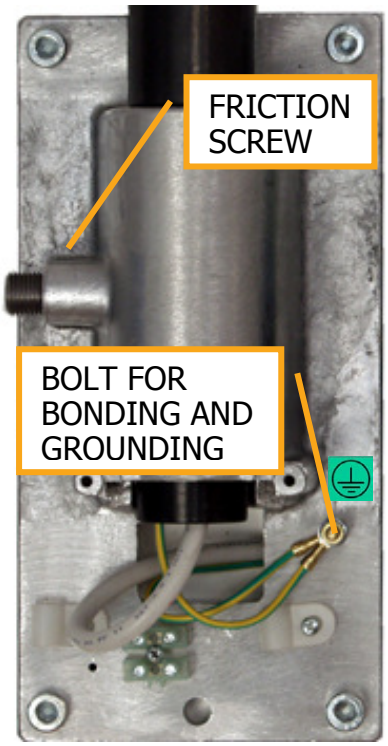
Maintenance for the FONA XDG system to be done regularly by a service technician at least once every 24 months, with regular checks performed by the operator every year.
Any defect or malfunction should be corrected immediately by qualified personnel with adequate training.

WARNING.

- Any defective item affecting A safe use must be repaired or replaced
- Only original certified components and spare parts must be used for repairs or replacements. Correction of damages to the identification labels to be handled with the Manufacturer. Any defect or non-compliance must be reported promptly to the Manufacturer or to its Local Agent.
- Always disconnect the system from the line voltage supply using the general line voltage switch for the room where the equipment is located before performing any maintenance activity
- Avoid using liquid or spray detergents which may enter into the equipment and cause corrosion.
- Avoid the use of solvents or corrosive detergents which can damage painted surfaces and plastic covers.

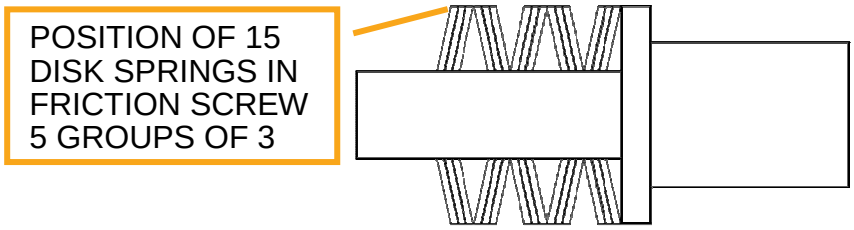
4.1 Maintenance of Wall Adaptor

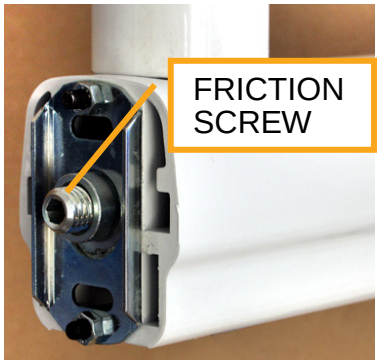




A Remove the cover and verify that the mounting is closely connected to the wall and stays firm and steady during the movements of the system.
B Verify for proper connection of grounding wires to the common bolt.
C Check friction during rotation of Extension Arm and adjust if needed.
D Verify that the technical label with identification data outside the plastic cover at bottom is in place and is readable.

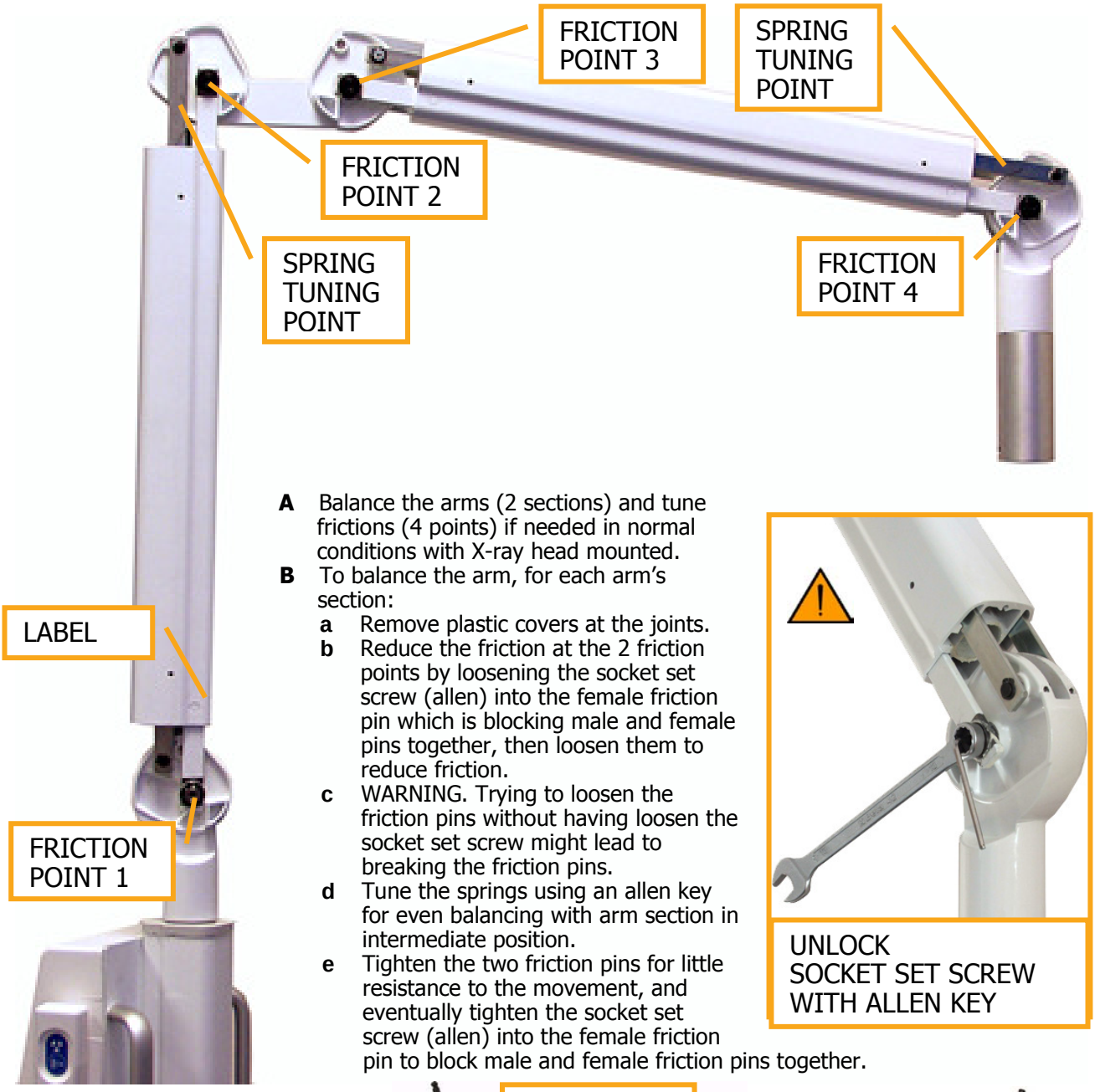
4.2 Maintenance of Support Arm





A Slide partially out the extension arm from wall mount, inspect for wear of the joint and lubricate for smooth rotation.
B Check friction during rotation of Arm and adjust relevant screw if needed.
C Verify that the technical label with identification data is in place and is readable.

4.3 Maintenance of Scissor Arm



A Balance the arms (2 sections) and tune frictions (4 points) if needed in normal conditions with X-ray head mounted.

B To balance the arm, for each arm's section:

- **a** Remove plastic covers at the joints.
- **b** Reduce the friction at the 2 friction points by loosening the socket set screw (allen) into the female friction pin which is blocking male and female pins together, then loosen them to reduce friction.
- **c** WARNING. Trying to loosen the friction pins without having loosen the socket set screw might lead to breaking the friction pins.
- **d** Tune the springs using an allen key for even balancing with arm section in intermediate position.
- **e** Tighten the two friction pins for little resistance to the movement, and eventually tighten the socket set screw (allen) into the female friction pin to block male and female friction pins together.

C Specifically for the arm, at least every 24 months in case of normal use or after every 5000 cycles performed, have the service technician inspect for wear of pins and levers.

- **a** Remove plastic covers.
- **b** Check for wearing out of pin and levers, starting from the joint at the base of the arm.
- **c** Make sure no part is loose and no vibrations are occurring during movements.
- **d** Apply silicon spray lubricant if needed.
- **e** Replace the worn out parts or the arm itself if they appear damaged.

D To replace defective pins or levers

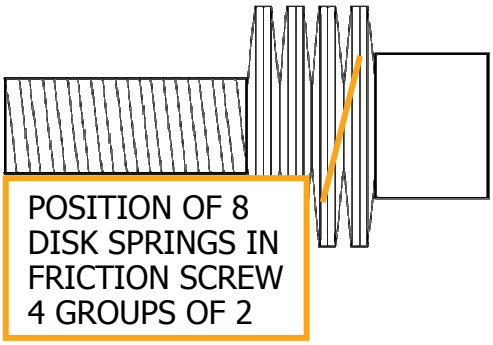
- **a** have the X-ray head disconnected with arm fully open to avoid injuries for sudden opening.
- **b** Loosen completely the two springs of the arm before extracting pins or removing levers.

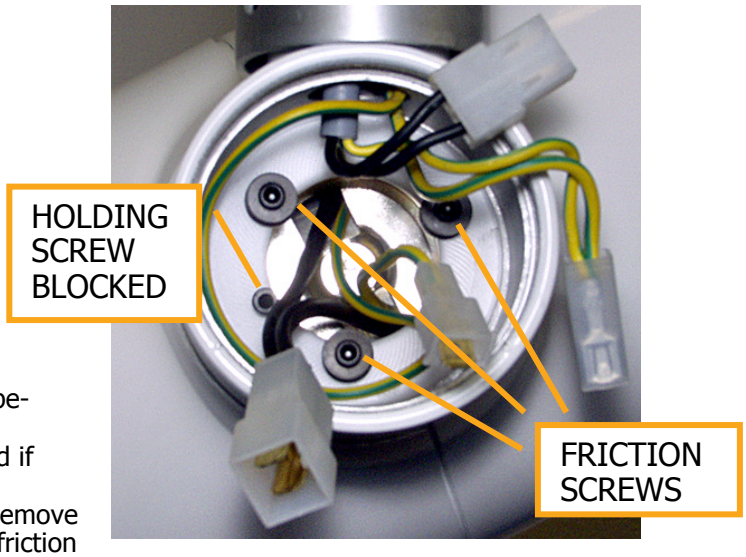
E Mount back the X-ray head if removed and secure it with the retaining fork.

F Slide partially out the Scissor Arm from the Support Arm, inspect for wear of the vertical joint and lubricate for smooth rotation.

G Verify that the label with identification data is in place and is readable.

4.4 Maintenance of X-ray Head





A Inspect for damage/wear to the X-ray tube-head and support system.
B Inspect for oil leakage. Replace tube-head if necessary.
C Check for position stability. If necessary remove the cap on the yoke side, tune the three friction screws (with disk springs) to be moderately tight for smooth rotation. The fourth one without disk springs must be blocked in place
D Check that electrical cables are correctly twisted and not damaged or worn down.
E Verify that the technical label with identification data is in place and is readable.
F Dismount the tube-head from the arm and lubricate the pin for smooth rotation. Warning: remove the tube-head with the arm completely open to avoid the risk of injuries.
G Check the retaining clip and replace if worn down. Do not mount the tube-head until all checks of the suspension system are completed.

4.5 Maintenance of Beam Limiting Device

A Verify that the collimator is properly mounted and firmly closed.
B Verify that the internal cone is not damaged. Replace the collimator if necessary.
C Verify that the technical label with identification data is in place and is readable.

4.6 Maintenance of Timer

A Inspect the panel for damages on the surface.
B Inspect the cable of the exposure switch for wear.
C Check that all lights (except the yellow one) are working when the system is switched on.
D Perform a functional check with verification of a) setting of dip switches, b) yellow light, c) buzzer, d) alarm A07 for "Exposure push button pressed at power on", e) alarm A08 for "Exposure stopped by the operator".
E Verify that the fuses mounted comply with those indicated on the label and listed below.
F Verify that the technical label with identification data is in place and is readable.

Fuses of FONA XDG Timer		
Ref.	115 V	230 V
F1	6.3 AT 5x20	4 AT 5x20
F2	6.3 AT 5x20	4 AT 5x20
F3	315 mAT 5x20	315 mAT 5x20
F4	80 mAT 5x20	50 mAT 5x20

4.7 Checking Filament Pre-Heating Time

During the life of the system the wearing down of the filament may result in a different pre-heating time (PHT). The value set at installation needs therefore to be verified and in case corrected.

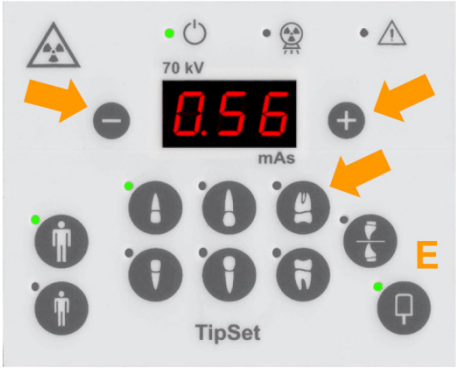
A At a fixed distance measure the X-ray dose for exposure of 2.0 s (nominal irradiation of 100 pulses at 50 Hz, 120 at 60 Hz).
B At same distance measure the dose for exposure of 0.2 s (nominal irradiation of 10 pulses at 50 Hz, 12 at 60 Hz).
C If dose at 0.2 s is 1/10 of dose at 2 s, the value set for PHT is fine.
D If dose at 0.2 s is lower than 1/10 of the dose at 2 s, PHT has to be increased. Proceed at steps of 20 ms until fine.
E If dose at 0.2 s is higher than 1/10 of the dose at 2 s, PHT has to be reduced. Proceed at steps of 20 ms until fine.

4.8 Service Functions

A READING EXPOSURE COUNTER. Enter by switching ON the unit while pressing at the same time for 2 s, the three keys plus (+), minu s (−), and upper incisor. The cumulative number of exposures is of 9 digits shown on the display in groups of three digits, from the most to the least significant, selectable using the plus (+), minus (−) keys. Each group is identified by the dot at bottom, on the left for the most significant group, in centre for the intermediate group, at the right for the least significant one. Press the sensor key to exit (E).

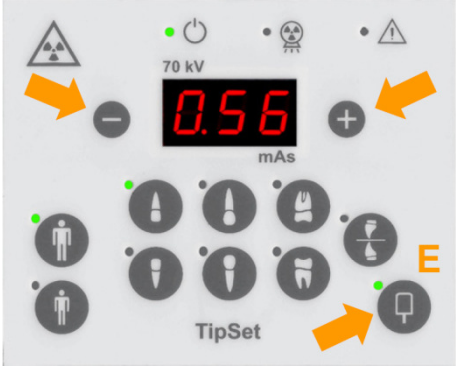
B READING ENERGY COUNTER. Enter by switching ON the unit while pressing at the same time for 2 s, the three keys plus (+), minus (−), and upper molar. The cumulative value of the energy imparted in Joule (J) is of 9 digits shown on the display in groups of three digits, from the most to the least significant, selectable using the plus (+), minus (−) keys. Each group is identified by the dot at bottom, on the left for the most significant group, in centre for the intermediate group, at the right for the least significant one.

Press the computer key to exit (E).



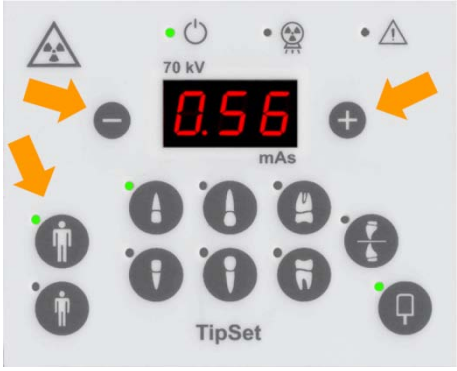
C DIGITAL VOLT METER. This function allows the user to display the line voltage level and the frequency, alternating.

- **a** Enter this function by switching ON the unit while pressing at the same time the (+), the (−), and the sensor keys.
- **b** Exit by switching the system OFF.



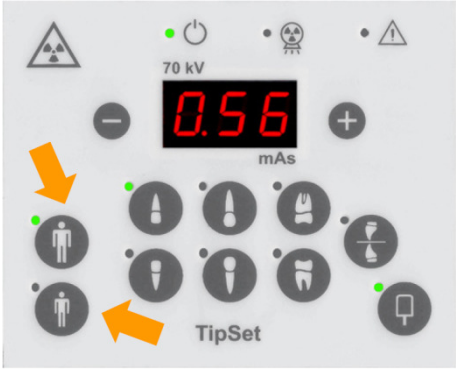
D TEST CONTROL PANEL. Function i to test lights, buttons, and segments of numerical display on the control panel.

- **a** Enter by switching the system "ON" while pressing at the same time the (+), the (−), and the adult keys.
- **b** Press the (+) or (−) keys to check the segments of the numerical display.
- **c** Press the X-ray pushbutton for the "rAY" message.
- **d** Press each button on the panel to turn ON and OFF the corresponding light.
- **e** At the last button pressed the system exit this function.



E FIRMWARE VERSION. This function returns the code of the version of firmware loaded.

- **a** Enter this function by switching the system "ON" while pressing at the same time the (+), the (−), and the adult child keys.
- **b** Exit by switching the system OFF.



5. MEASUREMENTS

All measurements to be done with the equipment supplied at nominal line voltage with specified line resistance (see section 2.1, System Supply, at page 5). The tolerances on the measured values have to take into account the precision of each measurement instrument. Measurements have to be done by trained personnel only to avoid the risk of electrical shock.

5.1 Line Voltage

The Line voltage can be measured using a Volt Meter for Alternate Current within the proper range. A service function can be activated on the timer to display the current line voltage in V and frequency in Hz (see section 4.8 of this Manual).

5.2 Anode Voltage – KVp

The kVp level is the actual peak value of the Anode voltage which stabilizes once the preheating time of the filament has elapsed (in about 0.2 s) and the high voltage transformer is actually loaded. The kVp level can be measured with a non-invasive kVp meter placed in front of the BLD following the instructions in the user manual of the instrument. A correct measure can be done with an exposure time of 500 ms or more having introduced a delay of the reading instrument of about 300 ms to allow the voltage level to stabilize after the pre-heating time has elapsed.

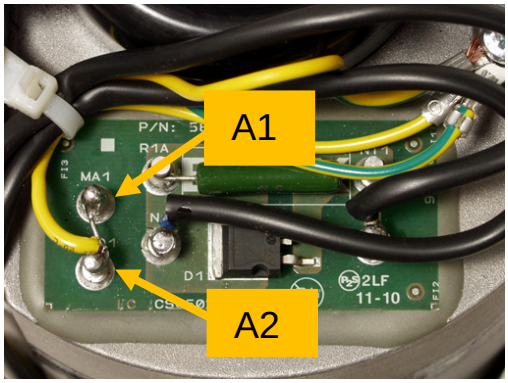
5.3 Anode Current – mA

WARNING. To prevent high voltage shock make sure the system is disconnected from the power supply when connections to the measurement points are performed. Electric discharge might occur in case of improper operation.

The Anode current in mA is the actual average value of the tube current which raises when the filament has warmed-up, after the pre-heating time has elapsed.

Remove the back cover of the tube-head, disconnect the jumper A1-A2 and replace it with a resistor of 1000 Ohm.

The anode current can be read with a DC voltmeter connected to points A1 and A2, with the 1 V corresponding to 1 mA. After the measurement disconnect the 1000 Ohm resistor and put back the jumper.



5.4 Exposure Time

Here below the timing indications in s and line pulses at 50 and 60 Hz for each value of exposure factor in mAs for tubehead at 3.5 mA.

mAs @3.5 mA	s	Pulses 50 Hz	Pulses 60 Hz
0.21	0,06	3	4
0.28	0,08	4	5
0.35	0,10	5	6
0.42	0,12	6	7
0.56	0,16	8	10
0.70	0,20	10	12
0.88	0,25	12	15
1.12	0,32	16	19
1.40	0,40	20	24

mAs @3.5 mA	s	Pulses @50 Hz	Pulses @60 Hz
1.75	0,50	25	30
2.24	0,64	32	38
2.80	0,80	40	48
3.50	1,00	50	60
4.40	1,25	62	75
5.60	1,60	80	96
7.00	2,00	100	120
8.75	2,50	125	150
11.2	3,20	160	192

The actual exposure (loading) time is "determined as the time interval between the instant when the tube current first rises above the 25% of its maximum value and the instant when it finally falls below the same value".

In order to assure the requested exposure time (irradiation), the pre-heating time of the insert used, which has been set-up at installation, is taken into consideration. The actual switch-on time (SOT) of the X-ray head assembly is therefore the sum of the filament pre-heating time (PHT) and of the requested exposure time (RET).

SOT = PHT + RET

The measurement of the exposure time can be done with kVp & Time meters with the ability to trigger at 75% of maximum kV level.

Exposure time can also be measured after having connected a resistor of 1000 Ohm between points A1 and A2 on the small circuit board on the back of the X-ray head, by monitoring the voltage across it (tube current) with an oscilloscope, and counting the number of pulses in the time interval.

When the function to correct the effects of line fluctuations on the emitted X-ray dose is activated, the exposure time is corrected in order to assure dose consistency. When the actual line voltage is lower than the nominal line voltage, the actual exposure time is increased. Conversely, when the actual line voltage is higher than the nominal line voltage, the actual exposure time is decreased.

Do disable this correction function (dip switch number 1 of SW2 moved to OFF position) to verify exposure time accuracy.

5.5 Leakage Radiation

The value of leakage radiation of the X-ray tube assembly is measured at 1 m distance taking into account the energy provided in 1 hour. In case the measurement is performed at a distance different from 1 m, the value has to be properly corrected.

The applicable technique factors for FONA XDG are of 70 kV and 3.5 mA. The point of maximum leakage has to be considered, excluding the direction of the primary beam. The duty cycle factor as reported in section 2.2, X-ray Head Assemblies, at page 5, has also to be taken into account for the computation of the actual dose rate. In case for example a measure is done at 50 cm distance from the source with an exposure of 1 s, the following computations have to be performed:

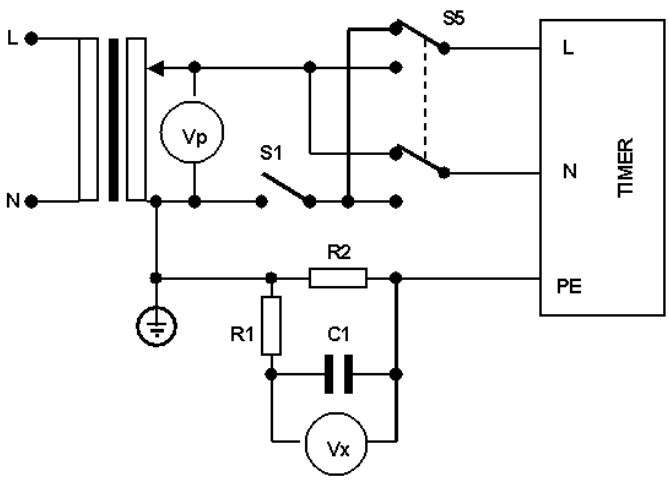
- A duty cycle of 1/15 means maximum of 1 s exposure every 15 s, i.e. 240 s per hour.
- Dose at 100 cm is computed taking into account the inverse square law, i.e. at twice the distance the dose intensity is reduced to one quarter. D1h100cm = D1s50cm * 240 * $(50/100)^2$ = D1s50cm * 60
  where: D1s50cm is the dose measured at 50 cm with 1 s exposure
  and D1h100cm is the dose computed at 1 m distance for the energy of 1 hour.

5.6 Earth Resistance-

The measurement of the earth resistance to be done after having disconnected the system for the power supply line. Using a voltage generator of not more than 6 V, 50/60 Hz, a current of 25 A is applied from 5 s to 10 s through the earth terminal in the connecting block of the timer to any metal part which can be in touch to the patient/operator (applied part). The voltage drop is then measured and the earth resistance computed. The system complies for values up to 0,1 Ω (extended to 0,2 Ω in case of mobile system).

5.7 Earth Leakage

Connect the Timer to the measuring circuit as shown in the figure (R1=10 kΩ ± 5%, R2=1 kΩ ± 1%, C1=0,015 F ±5%)



A Set the input supply voltage (Vp) equal to 110% of the nominal supply voltage.
B Connect a voltmeter (Vx with 1 MOhm input impedance) to the indicated points
C Switch the Timer "ON"
D Make two measures in Normal Conditions (NC with S1 "ON")
- S5 "ON": live and neutral conductors in normal condition.
- S5 "OFF": swap of live and neutral conductors.

E Make two measures in Single Fault Condition (SFC with S1 "OFF").
- S5 "ON": live and neutral conductors in normal condition.
- S5 "OFF": swap of live and neutral conductors.

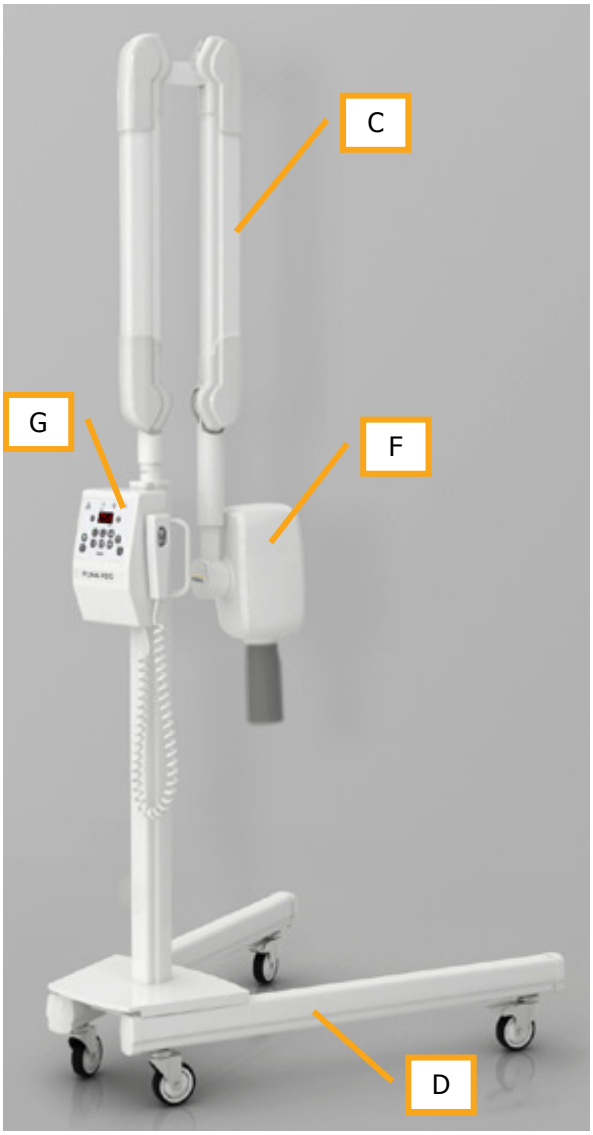
Please note that 1V equals to 1mA. Values of leakage currents have to be within the limits of table below.

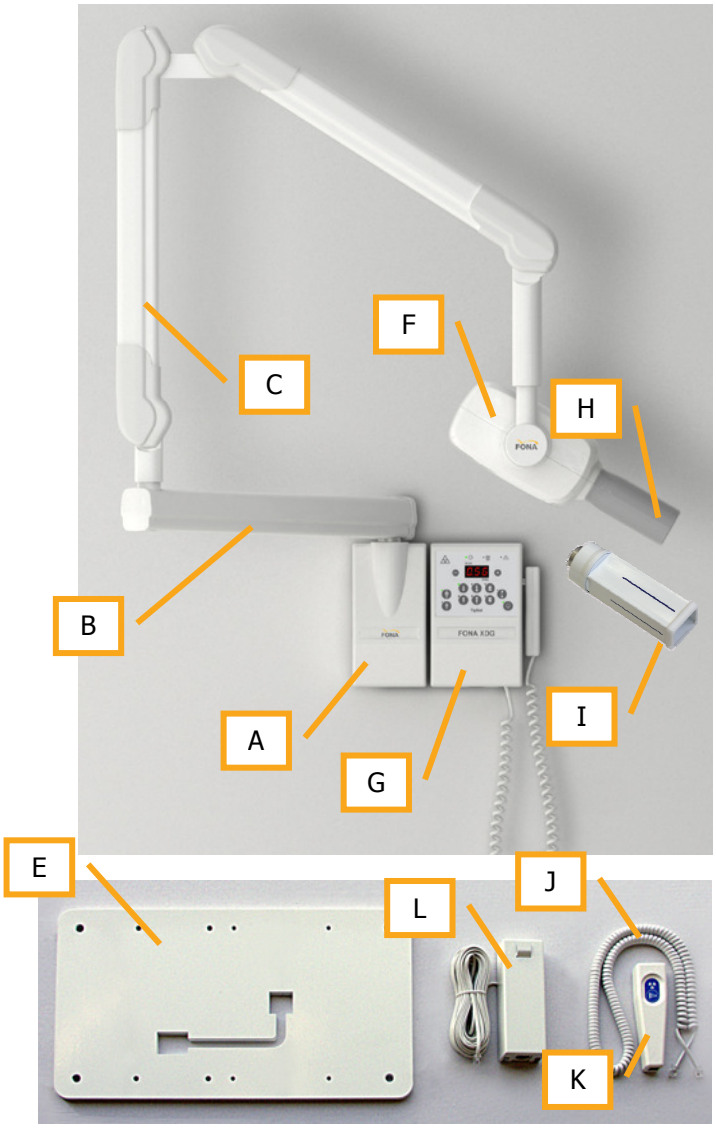
Limit of earth leakage current for X-ray equipment		
Type of connection to line voltage supply	Normal Condition (S1 "ON")	Single Fault Condition (S1 "OFF").
Permanent	5 mA	10 mA
With power cord	2,5 mA	5 mA

Appendix A
Systems and Components

FONA XDG Systems @ 230 V	
Code	Description
97 586 62320	Wall Syst. Arm S 30 cm 230V
97 586 62620	Wall Syst. Arm M 60 cm 230V
97 586 62820	Wall Syst. Arm L 80 cm 230V
97 586 62120	Wall Syst. Arm L 100 cm 230V
97 586 67820	Mobile System 230V

FONA XDG Systems @ 120 V	
Code	Description
97 546 62320	Wall Syst. Arm S 30 cm 120V
97 546 62620	Wall Syst. Arm M 60 cm 120V
97 546 62820	Wall Syst. Arm L 80 cm 120V
97 546 62120	Wall Syst. Arm L 100 cm 120V
97 546 67830	Mobile System 120V UL/CSA





ID	Description	Code
A	Wall adaptor	93 105 11000
B	Support arm S 30 cm	93 150 17100
	Support arm M 60 cm	93 150 17200
	Support arm L 80 cm	93 150 17300
	Support arm L 100 cm	93 150 17400
C	Scissor arm	93 150 12010
D	Mobile stand	93 150 20080
	Mobile stand UL/CSA	93 150 20090
E	Wall plate 16”	86 100 11500

ID	Description	Code
F	X-ray head XDG 120 V	93 256 01300
	X-ray head XDG 230 V	93 256 01700
G	TipSet Timer 110-120 V	93 356 60200
	TipSet Timer 220-240 V	93 356 60100
H	6 cm circular field BLD Grey	91 360 00020
I	3x4 cm X-ray field BLD	91 300 00040
J	Coiled cable	76 190 25580
K	Hand-switch W/O cable	76 190 25590
L	Kit remote hand-switch	93 300 02000

Appendix B
Icons

	IEC Type B Equipment		Compliance to European Community Requirements
	X-ray On Irradiation		Compliance to Canadian and US Standards
	Examine Annexed Documentation		Line voltage supply On - System Ready
	Increase Exposure Time (one step)		Off (Disconnected from Line voltage Supply)
	Decrease Exposure Time (one step)		On (Connected to Line voltage Supply)
	Child – Small Patient	~	Alternate Current
	Adult – Large patient		Fuse
	Upper Incisor		Protective Earth
	Upper Canine/Premolar	N	Neutral Point (for equipment permanent connected to line)
	Upper Molar	L	Live Point (for equipment permanent connected to line)
	Lower Incisor		Inherent Filtration
	Lower Canine/Premolar	■	Focal Spot
	Lower Molar		Fragile, Handle With Care
	Bite Wing - Interproximal		Fear of Humidity
	Digital Receptor		Up Do Not Overturn
	Radiography Push Button	3	Stacking Limit
	Ionizing Radiation		Separate Collection, Do Not Dispose

Appendix C
Exposure Table

	Large Patient ↓		Lower Incisor ↓	Upper Incisor / Lower Canine/ Premolar ↓	Upper Canine/ Premolar / Lower Molar or Bitewing	Upper Molar ↓	Small Patient ↓	
			0.21	0.28	0.35	0.42	← 0.32	
			0.28	0.35	0.42	0.56	← 0.40	CDR SENSOR
	0.32	→	0.35	0.42	0.56	0.70	← 0.50	XIOS SENSOR
CDR SENSOR	0.40	→	0.42	0.56	0.70	0.88	← 0.64	
XIOS SENSOR	0.50	→	0.56	0.70	0.88	1.12	← 0.80	
	0.64	→	0.70	0.88	1.12	1.40	← 1.00	E FILM
	0.80	→	0.88	1.12	1.40	1.75	← 1.25	
E FILM	1.00	→	1.12	1.40	1.75	2.24	← 1.60	
	1.25	→	1.40	1.75	2.24	2.80	← 2.00	D FILM
	1.60	→	1.75	2.24	2.80	3.50	← 2.50	
D FILM	2.00	→	2.24	2.80	3.50	4.40	← 3.20	
	2.50	→	2.80	3.50	4.40	5.60	← 4.00	
	3.20	→	3.50	4.40	5.60	7.00	EXPOSURE INDEX	
	4.00	→	4.40	5.60	7.00	8.75		
	EXPOSURE INDEX		5.60	7.00	8.75	11.20		
			3.50	4.40	5.60	7.00		
			4.40	5.60	7.00	8.75		
			5.60	7.00	8.75	11.20		

FONA XDG Exposure Factors in mAs

CDR Sensor: Schick Technologies. Inc. Long Island City, NY, USA
XIOS Sensor: Sirona Dental Systems GmbH, Bensheim, Germany
Film D: Kodak Ultraspeed, Agfa Dentus M2 – Film E: Kodak Ektaspeed Plus - Film F: Kodak Insight

Appendix D
Alarm Conditions

Timer Alarm Conditions			
Code	Fault /Error	Signal	Reset
A 01	X-ray requested during cool-down period	Green lamp (System Ready) flashing	By acknowledgement on the panel or when system has cooled down
A 02	Line voltage below lower limit	Green lamp (System Ready) and red lamp (Alarm) flashing	Automatically when line voltage is back in range
A 03	Line voltage above upper limit	Green lamp (System Ready) and red lamp (Alarm) flashing	Automatically when line voltage is back in range
A 06	Line Frequency Detection Failure	System Ready (green) lamp and Alarm (red) lamp flashing	By switching system off and on again
A 07	Exposure push button pressed at power on	Red lamp (Alarm) flashing	By acknowledgement on the panel
A 08	Exposure stopped by the operator	Red lamp (Alarm) flashing	By acknowledgement on the panel or after 1 m
A 09	Exposure stopped by the back-up timer	Red lamp (Alarm) switched on	By switching system off and on again
A 10	Back-up relay failure	Red lamp (Alarm) switched on	By switching system off and on again
A 11	Power switching device failure	Red lamp (Alarm) switched on	By switching system off and on again
A12	Line dips during exposure	Red lamp (Alarm) switched on	By acknowledgement on the panel

Appendix E
Identification Labels

FONA XDG 70 kVp 3.5 mA
120 V ~ 50/60 Hz 6 A TUBE CF4G070
0.2 kW 2.5 AI/70 0.4 EN 60336
TYPE 9325601300 SN 2701FG0001
MANUFACTURED DECEMBER 2010
COMPLIES WITH DHHS PERFORMANCE STANDARD 21 CFR SUBCHAPER J
FONA SRL
VIA IDIOMI 1/8-33 ASSAGO ITALY

FONA XDG 70 kVp 3.5 mA
230 V ~ 50/60 Hz 4 A TUBE CF4G070
0.2 kW 2.5 AI/70 0.4 EN 60336
TYPE 9325601700 SN 2701FF0001
MANUFACTURED JANUARY 2011
COMPLIES WITH DHHS PERFORMANCE STANDARD 21 CFR SUBCHAPER J
FONA SRL
VIA IDIOMI 1/8-33 ASSAGO ITALY

SCISSOR ARM
TYPE 9315012010
SN 2612GM00001

SUPPORT ARM 60
TYPE 9315017200
MODEL 6280239
SN 2412600001

TYPE 9130000020
SN 2612BY0123
DATE OF MANUFACTURE
DECEMBER 2010
FONA SRL
VIA IDIOMI 1/8-33
ASSAGO ITALY

WALL ADAPTOR
TYPE 9310511000
SN 2612WF00001

THIS PRODUCT COMPLIES WITH DHHS REGULATIONS 21 SUBCHAPTER J APPLICABLE AT DATE OF MANUFACTURE

WARNING:
THIS X-RAY UNIT MAY BE DANGEROUS TO THE PATIENT AND OPERATOR UNLESS SAFE EXPOSURE FACTORS AND OPERATING INSTRUCTIONS ARE OBESERVED. ELECTRICAL SHOCK HAZARD – DO NOT REMOVE PANELS. RISK OF EXPLOSION – DO NOT USE IN PRESENCE OF FLAMMABLE ANESTHETICS. FOR CONTINUED PROTECTION AGAINST RISK OF FIRE, REPLACE ONLY WITH SAME TYPE AND RATING FUSE.

COMPLIES WITH DHHS PERFORMANCE STANDARD 21 CFR SUBCHAPTER J

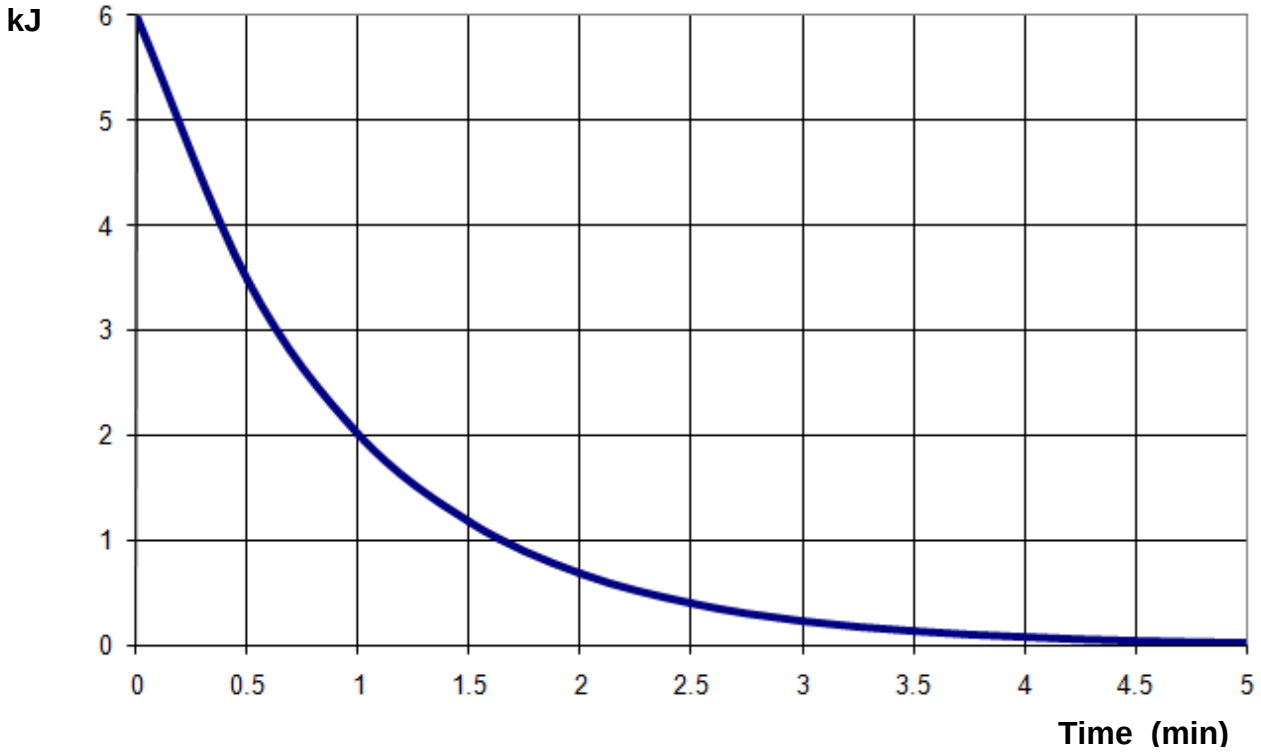
TipSet TIMER
0.06 – 3.2 s
MANUFACTURED DECEMBER 2010
TYPE 9335660200
SN 2612XL0199
110-120 V ~ 50/60 Hz
Fuse T 6.3 A
FONASRL
VIA IDIOMI 1/8-33
ASSAGO ITALY

TipSet TIMER
0.06 – 3.2 s
TYPE 9335660100
SN 2612XH0199
220-240 V ~ 50/60 Hz
Fuse T 4A
FONA SRL
VIA IDIOMI 1/8-33
ASSAGO ITALY

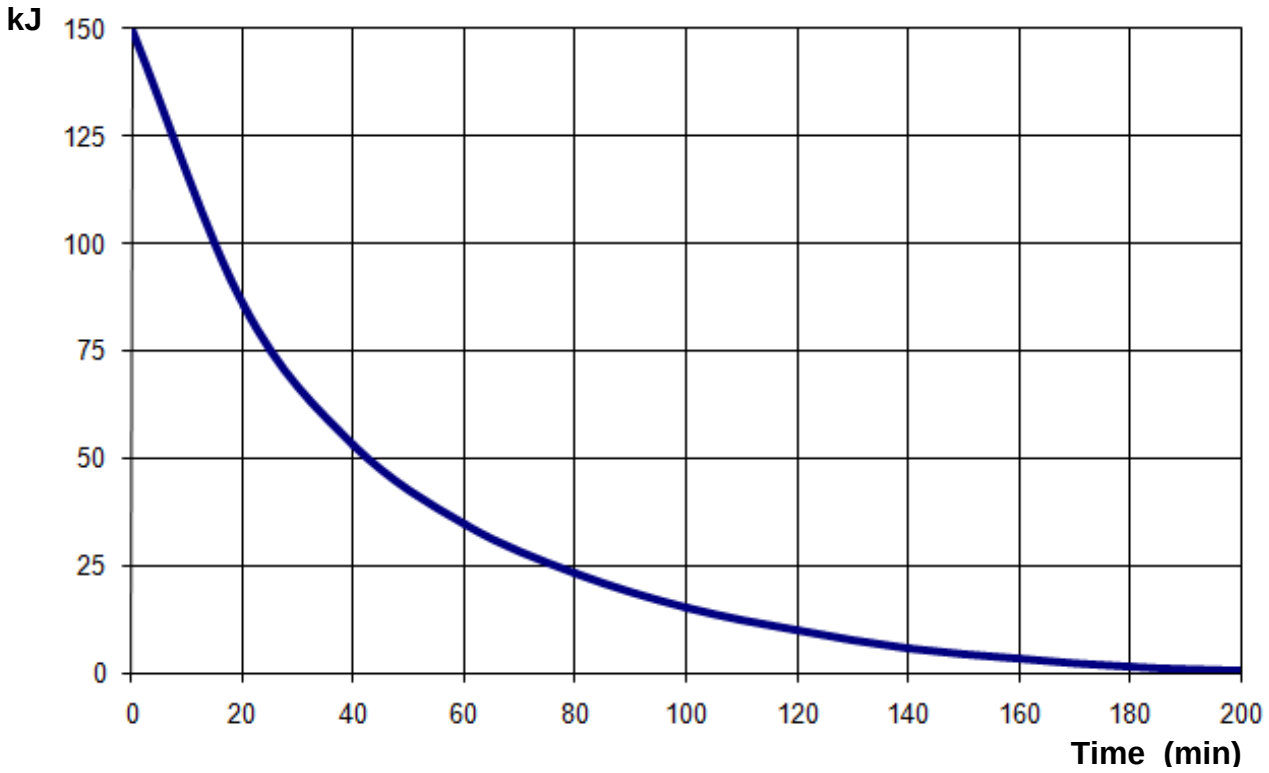
MOBILE STAND
TYPE 9315020080
MODEL 6280312
SN 2310GB0022

CLOSE FOR TRANSPORT
CHIUDERE PER TRAPORTO
FERMER POUR TRANSPORT
CIÉRRESE PARA TRANSPORTE
SIE WIRKLICH FÜR TRANSPORT

COOLING CURVE OF X-RAY INSERT



COOLING CURVE OF TUBE HOUSING ASSEMBLY



We reserve the right to make any alterations which may be required due to clinical improvements

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